Fair and Representative Clustering for Data Summarization

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Abstract

Clustering partitions the data into k clusters, and selects one center (or representative) from each cluster to form a summary. In *fair* clustering, each point belongs to one of t demographic groups, and existing algorithmic fairness notions seek a minimum number of representatives from each group.

In this work, we argue that existing criteria can lead to *tokenistic* summaries, in which centers mostly represent members of other groups. To address this problem, we propose a more expressive algorithmic fairness notion that captures how well each group is represented by centers from the same group, and generalizes prior work. We present a post-processing algorithmic framework to study this notion under the k-Supplier and k-Means clustering objectives. In particular, given a clustering of cost Γ and parameters $\lambda, \varepsilon > 0$, our approximation algorithm guarantees that each group receives at least $\mathrm{OPT} - \varepsilon k$ representation within clustering cost $\Gamma + \lambda$ for k-supplier, and $(1 + \lambda)\Gamma$ for k-means. Here, OPT denotes the optimal minimum group representation attainable within the stated clustering cost bounds.

Experiments on six real-world datasets show that on average, our approximation algorithm improves the representation of the most disadvantaged group by 2–3× and 4–5× under the k-Means and k-Supplier objectives respectively.

Keywords

Clustering, Algorithmic Fairness, Responsible Computing

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1 Introduction

Data summarization aims to provide a small, representative subset of a dataset, and is a fundamental task in many web systems [6, 9, 10, 18, 20, 33, 39], including digital participation platforms that summarize public input. Online platforms such as Pol.is, Talk to the City, Go Vocal and Remesh support many governments and civic organizations in collecting and understanding public input to inform policy decisions [8, 16, 31, 37]. Therefore, it is important that the data summaries produced by such platforms are representative of stakeholders from different demographic groups.

We study this problem in the following setting: A government seeks to summarize n responses for m questions, submitted by participants belonging to t demographic groups. How can they pick $k \ll n$ responses to best represent interests of the t groups? A common approach involves clustering the n responses and returning the k cluster centers as a data summary. To ensure representation for the t groups, prior work on fair clustering selects a given number of centers from each group [12, 29, 35].

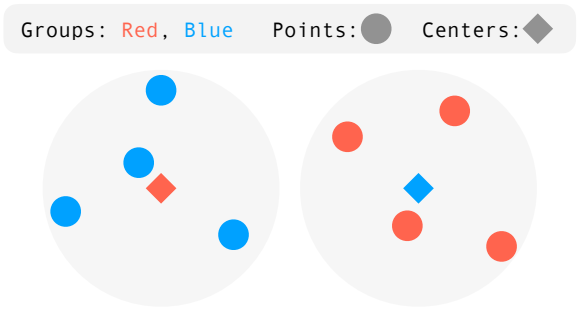


Figure 1: A data summary with $k = 2$ clusters. The data summary is fair in the sense that both groups have an equal number of centers but each center represents more points from the other group than its own, which is *tokenistic*.

However, only constraining the number of centers per group may lead to selecting arbitrary responses from each group. Particularly, it does not ensure that the selected responses reflect the views of the respective group, and for instance, the cluster represented by a response may consist predominantly of other groups' responses. Consider the data summary illustrated in Figure 1. Although it selects one center from each group, only 20% points from each group are represented by a center from the same group. Following this observation, we argue that one may cherry-pick responses to satisfy algorithmic fairness criteria based on center counts. We view such summaries as a form of *tokenism*, i.e., symbolic inclusion that denies groups meaningful representation [26].

Instead, replacing both centers in the illustration with any other point in their respective clusters would ensure that 80% points from each group are represented by a center of the same group.

Our Contributions. In this work, we introduce the *representative fair k-clustering* problem, which requires the selected centers to provide a given amount of representation to each group. To continue our data summarization example from above, this ensures that the responses selected by our algorithms are not tokenistic; instead, they are representative of their respective groups and clusters.

In our novel problem, we are given a set of points X, where each point belongs to one of t groups, and a representation constraint $\alpha \geq 0$. The goal is to partition X into k clusters $C_1, \ldots, C_k$ and select a center s_i for each cluster such that every group receives at least α representation, while minimizing the clustering cost.

We consider two notions of representation. In the *global* variant, we will ensure that at least α-fraction of each group's members are assigned to same-group centers. We also introduce a *local* variant, which ensures that each group receives at least α clusters' worth

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of representation, accounting for how well each center represents its cluster. See §3 for the formal definitions of the problem and the two notions of representation.

Further, we show that our novel problem contains the previously studied fair k-clustering problem [12, 29, 35] as a special case.

From an algorithmic perspective, we introduce two algorithms that obtain a clustering $C_1, \ldots, C_k$ of cost Γ and error parameters $\lambda, \varepsilon > 0$ as input. They return a clustering of cost $(1+\lambda)\Gamma$ in case of k-Means clustering and $\Gamma + \lambda$ in case of k-Supplier clustering, while ensuring that each group has representation at least $\mathrm{OPT} - \varepsilon k$. Here, OPT is the optimal minimum group representation that is achievable within the bounds on the clustering cost defined above.

The first algorithm rounds the representation values of the clusters for each group, allowing us to find an approximately optimal assignment of clusters to groups using an integer linear program. The second algorithm uses randomized rounding of a linear program to achieve a similar guarantee with high probability.

Experiments on six real-world datasets show that our algorithms improve the representation of the most disadvantaged group by a factor of 2–3× for k-Means and 4–5× for k-Supplier, compared with previous cardinality-based algorithms [18, 35]. We further find that our methods are highly scalable, finishing within less than 20 minutes on datasets with up to 10 million points.

Our code is available online in an anonymized repository [3].

2 Preliminaries

We write $[n]$ to denote the set $\{1, \ldots, n\}$. Consider a metric space (X, d) with $X \subseteq \mathbb{R}^m$, $|X| = n$ and distance function $d : X \times X \to \mathbb{R}_{\geq 0}$. A k-clustering (C, S) is a partition of X into k clusters $C = \{C_i\}_{i\in[k]}$ and associated centers $S = \{s_i\}_{i\in[k]} \subseteq X$, where s_i is the center of C_i. We use $\mathrm{cost}(C, S)$ to denote the clustering cost.

In k-Supplier, the centers can only be chosen from a given set of facilities $F \subseteq X$, whereas in k-Center, they can be chosen from X. Both problems minimize the egalitarian objective:

$$\mathrm{cost}_\infty(C, S) = \max_{i\in[k]} \max_{x\in C_i} d(x, s_i).$$

In k-Means, the objective is to minimize the sum of squared distances between points and their corresponding cluster centers:

$$\mathrm{cost}_2(C, S) = \sum_{i\in[k]} \sum_{x\in C_i} \|x - s_i\|_2^2.$$

We are interested in constrained k-Means where $S \subseteq X$.

Let $\mathcal{G} = \{G_j\}_{j\in[t]}$ be a partition of X into t groups. We now provide an overview of how prior work conceptualizes *fair* clustering definitions with respect to these t groups. More precisely, for both k-Supplier and k-Means, a Fair k-Clustering requires that at least β_j centers belong to G_j; we formalize this definition below.

Definition 1 (Fair k-Clustering [12, 29, 35]). *Given a requirement vector $\vec{\beta} = (\beta_j)_{j\in[t]} \in \mathbb{R}^t_{\geq 0}$, a set of k clusters C with facilities $S \subseteq F$ is a* Fair k-Clustering *if it satisfies*

$$|S \cap G_j| \geq \beta_j$$

for all $j \in [t]$. The objective is to find a Fair k-Clustering (C, S) *that minimizes the clustering cost.*

The choice of clustering objective yields two variants of this problem, both of which have been studied in the literature:

(1) $\mathrm{cost}_\infty(C, S)$: Fair k-Supplier [12, 29]
(2) $\mathrm{cost}_2(C, S)$: Div k-Means [35]

All missing proofs are deferred to the appendix.

3 Fair Clustering with Representation

In this section, we first present a more expressive algorithmic fairness criterion for data summarization, that captures how well each center represents its clusters, and show that it generalizes prior definitions for Fair k-Clustering.

Definition 2 (α-Representative Fair k-Clustering). *Given a minimum representation constraint $\alpha \in \mathbb{R}_{\geq 0}$, a set of k clusters C with centers S is an* α-Representative Fair k-Clustering, *if it satisfies*

$$\mathrm{minrep}(C, S) := \min_{G\in\mathcal{G}} \sum_{i\in[k]} \mathbf{1}(s_i \in G)\, \mathrm{r}(C_i, G) \geq \alpha.$$

Here, $\mathrm{r} : 2^X \times \mathcal{G} \to [0, 1]$ is the representation function, *and $\mathbf{1}(s_i \in G) = 1$ if $s_i \in G$ and $\mathbf{1}(s_h \in G) = 0$ otherwise.*

The objective is to find an α-Representative Fair k-Clustering *that minimizes* $\mathrm{cost}(C, S)$.

In our work, we consider two variants of the representation function r:

$$\mathrm{r}_{\mathrm{global}}(C, G) := \frac{|C \cap G|}{|G|} \quad \text{and} \quad \mathrm{r}_{\mathrm{local}}(C, G) := \frac{|C \cap G|}{|C|},$$

where $\mathrm{r}_{\mathrm{global}}(C, G)$ denotes the fraction of group G contained in cluster C, whereas $\mathrm{r}_{\mathrm{local}}(C, G)$ denotes what fraction of cluster C is in group G. As shorthand, we use $\mathrm{minrep}_{\mathrm{global}}$ when it uses $\mathrm{r}_{\mathrm{global}}$, and $\mathrm{minrep}_{\mathrm{local}}$ for it uses $\mathrm{r}_{\mathrm{local}}$. Below, we elaborate on why we invoke this terminology.

For $\mathrm{minrep}_{\mathrm{global}}$, the constraint requires at least an α-proportion of each group's members to be represented by same-group centers across all clusters (hence, *globally*). Note that $\alpha_{\mathrm{global}} \in [0, 1]$.

In $\mathrm{minrep}_{\mathrm{local}}$, each group's representation is a weighted count of its centers, where each center is weighted by the fraction of the cluster belonging to its group. Thus, $\mathrm{r}_{\mathrm{local}}$ ensures that we pick centers from each group that represent their clusters well — which is a *local* property that for each center only depends on the respective cluster. Since each center contributes at most one unit to α_{local}, and at least one group has at most $\lfloor k/t \rfloor$ centers, we have $\alpha_{\mathrm{local}} \in [0, \lfloor k/t \rfloor]$.

In the following we will say that a cluster C_i is *allocated* to a group G_j if its center s_i is a member of group G_j. Similarly, we say that the *utility* of a cluster C for a group G is given by $\mathrm{r}(C, G)$.

Next, let us discuss some properties of α-Representative Fair k-Clustering.

First, note that we obtain a 1-Representative Fair k-Clustering for $\mathrm{r}_{\mathrm{global}}$, when all points belong to the same group as their cluster centers. If, in addition, at least $\lfloor k/t \rfloor$ centers are selected from each group, we get a $\lfloor k/t \rfloor$-Representative Fair k-Clustering under $\mathrm{r}_{\mathrm{local}}$.

Second, in the following lemma we show that Fair k-Clustering is a special case of α-Representative Fair k-Clustering.

Lemma 3. *The* Fair k-Clustering *problem is a special case of the* α-Representative Fair k-Clustering *problem.*

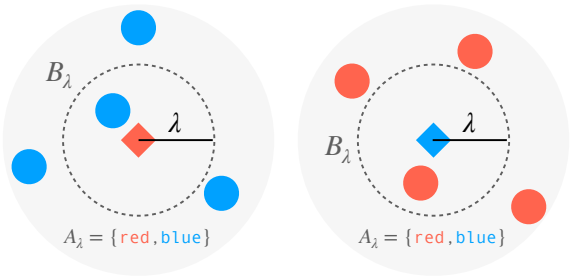


Figure 2: An illustration of how our algorithmic framework identifies candidate centers B_λ and corresponding groups A_λ.

4 Algorithmic Framework

We present our algorithmic framework to solve α-Representative Fair k-Clustering. Given an (unfair) clustering as input, it consists of a two-step post-processing procedure that returns a fair clustering (if one exists) while incurring a parameter dependent increase in the clustering cost. More precisely, given a k-clustering (C, S) of cost Γ, the first step identifies candidate centers that can replace the original center s_i of each cluster $C_i \in C$. In turn, these candidates restrict the groups to which each cluster can be allocated. Then, the second step computes an allocation of clusters to candidate groups that approximately maximizes the minimum (max-min) representation of any group.

To ensure that the resulting fair clustering does not incur a high clustering cost, our first step restricts center replacements so that the increase in the clustering cost is bounded as a function of some input parameter λ. The second step approximates the optimal max-min representation attainable with the selected candidates, with an additive error controlled by ε.

Next, we describe the two steps in more detail.

Step 1. For each cluster C_i, we use $B_\lambda(i)$ to denote the set of candidate centers that lie within distance (i) λ of its center s_i for k-Supplier, and (ii) $\sqrt{\lambda\Gamma_i/|C_i|}$ of its centroid s_i for k-Means, where $\Gamma_i = \sum_{x\in C_i} \|x - s_i\|_2^2$. For k-Supplier, the candidate set $B_\lambda(i)$ must also be a subset of the facility set F. Lemmas 4 and 5 bound the clustering cost for k-Supplier and k-Means respectively, when each original center s_i is replaced by a new center $s'_i \in B_\lambda(i)$.

Lemma 4. *Let (C, S) be a clustering with cost Γ on the k-Supplier objective, and let $S' = \{s'_i\}_{i\in[k]}$ such that $s'_i \in B_\lambda(i)$ for every $i \in [k]$. The clustering (C, S') has cost at most $\Gamma + \lambda$.*

The bound for k-Center follows as a corollary of Lemma 4.

Lemma 5. *Let (C, S) be a clustering with cost Γ on the k-Means objective, and let $S' = \{s'_i\}_{i\in[k]}$ such that $s'_i \in B_\lambda(i)$ for every $i \in [k]$. The clustering (C, S') has cost at most $(1 + \lambda)\Gamma$.*

Lastly, we define $A_\lambda(i) = \{G \in \mathcal{G} : G \cap B_\lambda(i) \neq \emptyset\}$ as the set of groups to which cluster C_i can be allocated.

Step 2. To maximize $\text{minrep}(C, S')$ for the new set of centers $S' = (s'_i)_{i\in[k]}$, we define the Fair Allocation problem:

Definition 6 (Fair Allocation). *An instance consists of a k-clustering (C, S), a representation function r, a set of candidate centers $(B_\lambda(i))_{i\in[k]}$, and corresponding groups $(A_\lambda(i))_{i\in[k]}$.*

The objective is to allocate each cluster C_i to a group G in $A_\lambda(i)$ by selecting a new center $s'_i \in B_\lambda(i) \cap G$ to maximize $\text{minrep}(C, S')$, where $S' = (s'_i)_{i\in[k]}$.

To ensure that the solution to Fair Allocation preserves the clustering cost bounds from *Step 1*, each cluster C_i is only allocated to a group in $A_\lambda(i)$, from which a center in $B_\lambda(i)$ can be selected. If $G \notin A_\lambda(i)$, then no center in $B_\lambda(i)$ belongs to G. Therefore, regardless of the value of $\mathrm{r}(C_i, G)$, every replacement center $s'_i \in B_\lambda(i)$ satisfies $\mathbf{1}(s'_i \in G)\,\mathrm{r}(C_i, G) = 0$. Accordingly, we simplify our Fair Allocation instance by setting $\mathrm{r}(C_i, G) = 0$ whenever $G \notin A_\lambda(i)$, and leaving it unchanged when $G \in A_\lambda(i)$, without changing the objective value of any feasible solution.

In the next two subsections, we present two algorithms for solving the Fair Allocation problem. More specifically, given some error parameter $\varepsilon \in (0, 1)$, our algorithms return an allocation with objective value at least $\text{OPT} - \varepsilon k$, where OPT denotes the optimal objective value.

4.1 Additive Approximation Scheme

Our algorithm is based on an ILP that allocates clusters $C_i \in C$ to groups $G \in A_\lambda(i)$ to solve the Fair Allocation problem. We start by providing a high-level overview of our approach before presenting the details.

First, to ensure that our ILP has a few variables, we round the representation function, reducing the number of distinct utilities from $O(tk)$ to a constant for fixed ε. Recall that t denotes the number of groups whereas the utility of a cluster for a group is given by the representation function. Then, clusters are binned into *profiles*, where each *profile* consists of clusters with identical rounded utilities across all groups. Second, we use these profiles to model the rounded Fair Allocation instance as an ILP.

In the first step, we incur an additive loss of at most εk in the objective value due to rounding. At the same time, for fixed t and ε, we obtain a constant number of cluster profiles. This allows us to model the rounded Fair Allocation instance as an ILP with a fixed number of variables, which is solvable exactly in polynomial time due to a result by Lenstra [30].

We describe each of these steps in more detail below.

Rounding. Given a parameter $\varepsilon \in (0, 1)$, we first set every utility below ε to zero and round each remaining utility down to the nearest power of $1 + \varepsilon$. Formally, we define

$$\mathrm{r}'(C, G) = \begin{cases} 0, & \mathrm{r}(C, G) < \varepsilon, \\ \max_{i\in\mathbb{N}_0}\{(1+\varepsilon)^{-i} : (1+\varepsilon)^{-i} \le \mathrm{r}(C, G)\}, & \text{otherwise.} \end{cases}$$

Since $\mathrm{r}(C, G) \in [0, 1]$, the rounded representation function r' takes at most $L_\epsilon = 2 + \lceil \log_{(1+\epsilon)}(1/\epsilon) \rceil$ distinct values. Next, for each cluster $C \in C$, we define its *profile* as $\mathbf{p} = (\mathrm{r}'(C, G_j))_{j\in[t]}$. Observe that if two clusters share the same rounded utility for each of the t groups, they also have the same profile. We use $k_\mathbf{p}$ to denote the number of clusters with profile $\mathbf{p}$, and $\mathcal{P}$ to refer to the set of all distinct cluster profiles.

Profile ILP. To recap, we are given a set of distinct profiles $\mathcal{P}$ and their corresponding profile counts $(k_\mathbf{p})_{\mathbf{p}\in\mathcal{P}}$. For each cluster with profile $\mathbf{p}$, we write $\mathbf{p}_j$ to denote its utility for group j. Our ILP intends to allocate $y_{\mathbf{p},j}$ clusters with profile $\mathbf{p}$ to every group $j \in [t]$ to maximize the minimum representation R attained by

any group. Note that all clusters with the profile $\mathbf{p} = (0, \ldots, 0)$ are omitted from the ILP, and allocated separately, as we will describe later. Our ILP, which we call the Profile ILP, is as follows:

$$\max_{\mathbf{y}, R} \quad R \tag{1}$$

$$\text{subject to} \quad \sum_{\mathbf{p} \in \mathcal{P}} \mathbf{p}_j y_{\mathbf{p},j} \geq R \qquad \forall j \in [t], \tag{2}$$

$$\sum_{j \in [t]} y_{\mathbf{p},j} = k_{\mathbf{p}} \qquad \forall \mathbf{p} \in \mathcal{P}, \tag{3}$$

$$y_{\mathbf{p},j} \in \mathbb{Z}_{\geq 0} \qquad \forall \mathbf{p} \in \mathcal{P}, j \in [t]. \tag{4}$$

Constraint (2) ensures that the overall representation of every group is at least R, whereas (3) enforces that exactly $k_{\mathbf{p}}$ clusters with profile $\mathbf{p}$ are allocated across all groups.

Next, we describe how we recover a solution for Fair Allocation from the solution returned by the Profile ILP. Let $\mathbf{y} = (y_{\mathbf{p},j})_{\mathbf{p} \in \mathcal{P}, j \in [t]}$ denote the optimal Profile ILP solution with objective value ALG. We assign $y_{\mathbf{p},j}$ clusters with profile $\mathbf{p}$ to group j for every $\mathbf{p} \in \mathcal{P}$ and $j \in [t]$, choosing arbitrarily among clusters with the same profile. Each cluster C_i omitted from the ILP is simply assigned to the candidate group maximizing its original utility given by $\arg\max_{G \in A_\lambda(i)} \mathrm{r}(C_i, G)$. Finally, for each cluster C_i assigned to G, we select its new center s_i' from $B_\lambda(i) \cap G$. These centers, together with the original clusters, form the solution (C, S') returned by our additive approximation scheme.

Next, we state the theoretical guarantees of our algorithm. Let OPT denote the optimal objective value for an instance of the Fair Allocation problem.

Theorem 7. *For a fixed t, and any $\varepsilon \in (0, 1)$, there exists an algorithm with running time $f(1/\varepsilon, t)\,\mathrm{poly}(k, t)$ that returns a solution of value at least $\mathsf{OPT} - \varepsilon k$, where f is a computable function.*

Proof. First, we bound the loss in the objective due to rounding. For every cluster $C \in \mathcal{C}$, and group $G \in \mathcal{G}$, we have that $\mathrm{r}(C, G) - \mathrm{r}'(C, G) \leq \varepsilon$: if $\mathrm{r}(C, G) < \varepsilon$, then the claim follows immediately. Otherwise, rounding down to a power of $(1 + \varepsilon)$ implies that $\mathrm{r}(C, G) \leq (1 + \varepsilon)\,\mathrm{r}'(C, G)$. Since $\mathrm{r}'(C, G) \leq 1$, we have

$$\mathrm{r}(C, G) - \mathrm{r}'(C, G) \leq (1 + \varepsilon)\,\mathrm{r}'(C, G) - \mathrm{r}'(C, G) = \varepsilon\,\mathrm{r}'(C, G) \leq \varepsilon.$$

Now, let $S^* = \{s_i^*\}_{i \in [k]}$ denote the optimal set of centers for the Fair Allocation instance. For each $\mathbf{p} \in \mathcal{P}$ and $j \in [t]$, let $\hat{y}_{\mathbf{p},j}$ denote the number of clusters with rounded profile $\mathbf{p}$ whose centers in S^* belong to G_j. Then, the corresponding representation of group G_j in the rounded instance can be written as [1]

$$\begin{aligned}
\sum_{\mathbf{p} \in \mathcal{P}} \mathbf{p}_j \hat{y}_{\mathbf{p},j} &= \sum_{i \in [k]} \mathbf{1}(s_i^* \in G_j)\,\mathrm{r}'(C_i, G_j) \\
&\geq \sum_{i \in [k]} \mathbf{1}(s_i^* \in G_j)(\mathrm{r}(C_i, G_j) - \varepsilon) \\
&\geq \mathsf{OPT} - \varepsilon k.
\end{aligned}$$

[1] Note that omitting the clusters with $\mathbf{p} = (0, \ldots, 0)$ does not affect the equality since $\mathbf{p}_j = 0$ for all $j \in [t]$.

Therefore, $\hat{\mathbf{y}}$ is a feasible solution to the Profile ILP for $R = \mathsf{OPT} - \varepsilon k$ and comparing it with its optimal solution ALG, we have that

$$\mathsf{ALG} \geq \min_{j \in [t]} \sum_{\mathbf{p} \in \mathcal{P}} \mathbf{p}_j \hat{y}_{\mathbf{p},j} \geq \mathsf{OPT} - \epsilon k.$$

Finally, since $\mathrm{r} \geq 0$, allocating clusters omitted from the ILP to the candidate groups maximizing their original utility does not incur any further loss to ALG.

Next, we bound the running time of our algorithm. Since every profile has t coordinates, with each coordinate taking at most L_ϵ values, we have $|\mathcal{P}| \leq \min\{k, L_\epsilon^t\}$ where $L_\epsilon^t = \mathrm{O}((\log_{(1+\epsilon)}(1/\epsilon))^t)$. [2] Hence, the Profile ILP has at most $tL_\epsilon^t + 1$ integer variables, which only depends on ϵ and t. Therefore, it can be solved exactly in time $f(1/\epsilon, t)\,\mathrm{poly}(k, t)$ for some computable function f, due to a result by Lenstra [30]. □

Theorem 8. *Unless P = NP, there exists no algorithm with running time $\mathrm{poly}(1/\varepsilon, t, k)$ that returns a solution of value $\mathsf{OPT} \pm \epsilon k$ for the Fair Allocation problem.*

4.2 Assignment LP with Randomized Rounding

In this section, we present a randomized algorithm for solving the Fair Allocation problem, based on the assignment LP of Bezáková and Dani [7]. Specifically, we apply multiple independent randomized-rounding trials to the optimal LP solution. Selecting the best rounded solution allows us to obtain an additive approximation to the optimal integral solution with high probability.

Recall that we are given a k-clustering (C, S), a representation function r, candidate centers $(B_\lambda(i))_{i \in [k]}$, and groups $(A_\lambda(i))_{i \in [k]}$. The Assignment LP intends to allocate fraction y_{ij} of cluster i to group j to maximize the minimum representation R attained by any group. We use $\mathrm{r}_{ij} = \mathrm{r}(C_i, G_j)$ to denote the utility of cluster i for group j. The LP is as follows:

$$\max_{\mathbf{y}, R} \quad R \tag{5}$$

$$\text{subject to} \quad \sum_{i \in [k]} \mathrm{r}_{ij}\, y_{ij} \geq R \qquad \forall j \in A_\lambda(i), \tag{6}$$

$$\sum_{j \in A_\lambda(i)} y_{ij} = 1 \qquad \forall i \in [k], \tag{7}$$

$$y_{ij} \geq 0 \qquad \forall i \in [k], j \in A_\lambda(i). \tag{8}$$

Constraint (6) ensures that the overall representation of any group in the fractional solution is at least R, whereas (7) enforces that the sum of fractional allocations of each cluster is exactly one.

Upon solving the LP, we obtain $\mathbf{y} = (y_{i,j})_{i \in [k], j \in A_\lambda(i)}$ along with its optimal solution value $\mathsf{OPT}_{\mathsf{LP}}$. Then, we apply ℓ independent randomized rounding trials to $\mathbf{y}$, where each trial proceeds as follows. First, we independently assign each cluster $C_i \in \mathcal{C}$ to group $G_j \in \mathcal{G}$ with probability y_{ij}. Second, for each cluster C_i assigned to G_j, we select its new center from $B_\lambda(i) \cap G_j$.

If $S_q = \{s_i^q\}_{i \in [k]}$ denotes the centers picked for trial $q \in [\ell]$, its solution has objective value $R_q = \mathrm{minrep}(C, S_q)$. Finally, we return the solution with the largest objective value over the ℓ trials, i.e., $\mathsf{ALG}_\ell = \max_{q \in [\ell]} R_q$.

[2] The number of distinct profiles $|\mathcal{P}|$ can never exceed the number of clusters k.

Next, we provide the theoretical guarantees for our algorithm. Again, we use OPT to denote the optimal objective value for the Fair Allocation problem.

Theorem 9. *Let $\epsilon, \delta > 0$, and ALG_ℓ denote the largest objective value obtained over ℓ independent randomized-rounding trials applied to an optimal solution of the assignment LP. If*

$$2\epsilon^2 k > \log t \qquad \text{and} \qquad \ell \geq \left\lceil \frac{\log(1/\delta)}{2\epsilon^2 k - \log t} \right\rceil$$

then we have $\text{ALG}_\ell \geq \text{OPT} - \epsilon k$ with probability at least $1 - \delta$.

We note that this theorem only holds when ε is large enough (the first condition requires that $\varepsilon \geq \sqrt{\log t / 2k}$). However, in our experiments we will see that from a practical perspective, the algorithm still yields very good results even for small ε.

Proof. For every cluster $C_i \in C$ and group $G_j \in \mathcal{G}$, we define the random variables $Y_{ij} = \mathbf{1}(s'_i \in G_j)\, \mathrm{r}_{ij}$ to model the representation contributed by C_i to G_j, and $Z_j = \sum_{i\in[k]} Y_{ij}$ as the total representation of G_j, where the randomness is due to the rounding that we described above. For a single randomized-rounding trial applied to $\mathbf{y}$, we use the linearity of expectation and the LP constraints to obtain

$$\mathbb{E}[Z_j] = \sum_{i\in[k]} \mathbb{E}[Y_{ij}] = \sum_{i\in[k]} \Pr(s'_i \in G_j)\, \mathrm{r}_{ij} = \sum_{i\in[k]} \mathbf{y}_{ij}\, \mathrm{r}_{ij} \geq \text{OPT}_{\text{LP}}.$$

Since $\text{OPT}_{\text{LP}} \geq \text{OPT}$, we have $\mathbb{E}[Z_j] \geq \text{OPT}$ for all $j \in [t]$. Next, as $0 \leq Y_{ij} \leq \mathrm{r}_{ij}$, we use Hoeffding's inequality to bound the probability of the bad event $Z_j \leq \text{OPT} - \epsilon k$ for any $j \in [t]$:

$$\Pr(Z_j \leq \text{OPT} - \epsilon k) \leq \exp\left(\frac{-2\epsilon^2 k^2}{\sum_{i\in[k]} \mathrm{r}_{ij}^2}\right) \leq \exp(-2\epsilon^2 k)$$

since $\sum_{i\in[k]} \mathrm{r}_{ij}^2 \leq k$. Next, we use a union bound for the probability that the rounded solution value is less than $\text{OPT} - \epsilon k$:

$$\begin{aligned}\Pr\left(\min_{j\in[t]} Z_j \leq \text{OPT} - \epsilon k\right) &\leq \sum_{j\in[t]} \Pr(Z_j \leq \text{OPT} - \epsilon k) \\ &\leq t \exp(-2\epsilon^2 k)\end{aligned}$$

Since the ℓ rounding trials are independent, we obtain the following guarantee for ALG_ℓ:

$$\begin{aligned}\Pr(\text{ALG}_\ell \leq \text{OPT} - \epsilon k) &= \Pr\left(\bigwedge_{q\in[\ell]} R_q \leq \text{OPT} - \epsilon k\right) \\ &= \prod_{q\in[\ell]} \Pr(R_q \leq \text{OPT} - \epsilon k) \\ &\leq \prod_{q\in[\ell]} t \exp(-2\epsilon^2 k) \\ &\leq t^\ell \exp(-2\epsilon^2 k\ell)\end{aligned}$$

To bound this probability, we set $t^\ell \exp(-2\epsilon^2 k\ell) \leq \delta$ to get

$$\Pr(\text{ALG}_\ell \geq \text{OPT} - \epsilon k) \geq 1 - \delta$$

which holds for

$$2\epsilon^2 k > \log t \qquad \text{and} \qquad \ell \geq \left\lceil \frac{\log(1/\delta)}{2\epsilon^2 k - \log t} \right\rceil. \qquad \square$$

5 Experiments

In this section, we provide an empirical evaluation of our algorithms, Profile ILP (§4.1) and Assignment LP with Randomized Rounding (§4.2), in comparison with previous work. Our experiments seek to answer the following questions:

- (RQ1) How do our algorithms perform on the α-Representative Fair k-Clustering objective?
- (RQ2) How does optimizing global representation affect local representation, and vice-versa?
- (RQ3) How does the dataset size impact the running time and solution quality of our algorithms?

Next, we provide details about our experimental setup, including baselines, input parameters concerning our algorithmic framework, implementation, and datasets.

Baselines. For k-Supplier, we compare our algorithms against (i) the greedy 3-approximation by Hochbaum and Shmoys [24], which we call 3APX k-Supplier, and (ii) the 3-approximation for Fair k-Supplier by Gadekar et al. [18], which we call 3APX Fair k-Supplier and whose approximation guarantee is best possible unless P = NP. For k-Means, we compare our algorithms against the k-means++ algorithm by Arthur and Vassilvitskii [2], and a state-of-the-art heuristic without theoretical guarantees by Thejaswi et al. [35], which is based on LP-rounding and local-search; we denote this algorithm Heuristic Div k-Means. We note that while Thejaswi et al. [35] also provide an FPT algorithm, they report that it does not scale well with the input size (a single run for $n = 123$ takes over 4 hours). Therefore we omit it from comparison since it does not scale to the size of our datasets. To the best of our knowledge, there are no other practical algorithms with theoretical guarantees that solve Div k-Means for data summarization.

Recall that Fair k-Clustering, and by extension both Fair k-Supplier and Div k-Means need a requirement vector as input; we construct two types of requirement vectors:

(1) *Equitable* $\vec{\beta} = (\lfloor k/t \rfloor)_{j\in[t]}$: Each group receives exactly $\lfloor k/t \rfloor$ centers.
(2) *Proportional* $\vec{\beta} = (\lfloor k \cdot |G_j|/|X| \rfloor)_{j\in[t]}$: Each group receives centers proportional to its group size. The requirements are set so that each group receives at least one center in order to avoid a vacuous Fair Allocation objective.

Algorithm Parameters. To compute the input clustering for our algorithmic framework, we use (i) the farthest-first traversal that yields a 3-approximation for k-Supplier [24], and (ii) the k-means++ algorithm that returns a $O(\log k)$-approximation in expectation for k-Means [2]. Next, we set the input parameters for our algorithms.

For k-Supplier, we set $\lambda = \Gamma$, where $\Gamma = \text{cost}_\infty(C, S)$, to match the clustering cost computed by the 3-approximation for Fair k-Supplier [18]. For k-Means, we consider varying values of λ to examine how increasing the clustering cost affects the feasibility of recovering a fair clustering; unless stated otherwise we use $\lambda = 1$.

We evaluate both of our algorithms on global and local variants of minrep. For the Profile ILP, we set $\varepsilon = 0.01/k$ to incur an additive loss of at most 0.01 from the optimal max-min representation. For the Assignment LP, we experiment with varying number of randomized rounding trials ℓ; unless stated otherwise we use $\ell = 64$.



Figure 3: Illustration of a hard Fair Allocation instance under global representation; colors denote groups. Clusters $C_1, \ldots, C_{k-1}$ each contain one ■, where each ■ contains all points of its color, and ● denotes a single blue point. To achieve non-zero representation for every group, all $k-1$ non-blue groups must receive their unique cluster. This leaves only cluster C_k for the blue group, which represents exactly $1/k$ blue points, implying that OPT $= 1/k$.

Implementation Details. We implement our algorithms in `python` using `Gurobi`, `NumPy`, and `scikit-learn` [21, 23, 32], and conduct our experiments on a compute server equipped with two AMD EPYC 9124 processors (2×32 cores) and 512 GB of RAM. All scalability experiments use a single core without parallelization. We perform ten independent runs for each configuration.

Datasets. We consider six datasets: `Heart`, `Student`, `MovieLens`, `CreditCard`, `Bank`, and `Adult` [22, 28] (see Table 1 for an overview of their sizes n, dimensions m, and number of groups t). We preprocess all datasets by applying one-hot encoding to categorical attributes, and min-max scaling to numerical attributes. Demographic groups are determined by the **intersection** of protected attributes: [3]

(1) `Heart`, `Student`, `MovieLens` ($t = 4$): `age` ($\leq$, $>$ `median`) and `sex` (`male`, `female`).
(2) `CreditCard` ($t = 8$): `age` ($\leq$, $>$ `median`), `sex` (`male`, `female`), and `marital-status` (`married`, `unmarried`).
(3) `Bank` ($t = 6$): `age` ($\leq$, $>$ `median`), `marital-status` (`single`, `married`, `divorced`).
(4) `Adult` ($t = 16$): `age` ($\leq$, $>$ `median`), `sex` (`male`, `female`), `race` (`White`, `Other`), and `native country` (`USA`, `Other`).

5.1 Algorithmic Fairness

We now present the algorithmic fairness evaluation of the baselines and our algorithms. For each dataset, we report minrep(C, S') for $\mathrm{r}_{\text{global}}$ and $\mathrm{r}_{\text{local}}$ on the k-clusterings computed by all algorithms for $k \in \{2t, 4t, 6t, 8t, 10t\}$. Due to space constraints, we only cover the results for k-Means in this section, and defer to Appendix §B.2 for the results on k-Supplier. Results for $k \in \{4t, 6t, 8t\}$ are included in the full version which is available in our anonymized repository [3].

Tables 1 and 2 report minrep(C, S') for the global and local representation functions respectively. First, we note that the Profile ILP outperforms all other algorithms on all datasets, as well as both representation objectives. Second, while the Div k-Means heuristic [35] offers substantial gains over k-means++, its still performs worse than the Assignment LP, with the exception of $k = 2t$ on `Adult` for the local objective. Third, the performance gap between the Assignment LP, and the Profile ILP is generally smaller for the local objective, and for larger values of k/t. We observe similar trends for how the Profile ILP and Assignment LP perform in comparison to the 3APX Fair k-Supplier by Gadekar et al. [18] (see Appendix §B.2).

[3] It is a significant limitation that these datasets equate `sex` with `gender`, and lack fine-grained labels for both attributes.

Next, averaged over all values of $k \in \{2t, 4t, 6t, 8t, 10t\}$ across the six datasets, the Profile ILP improves the max-min representation by a factor of about 2 for $\mathrm{r}_{\text{global}}$ and 3 for $\mathrm{r}_{\text{local}}$ when compared with Heuristic Div k-Means [35], and by a factor of about 5 for $\mathrm{r}_{\text{global}}$ and 4 for $\mathrm{r}_{\text{local}}$ when compared with 3APX Fair k-Supplier [18].

Thus, for **RQ1** we conclude that the Profile ILP is the best method to ensure fair representation of the groups.

Next, we closely examine `CreditCard` ($k = 2t$) which is the hardest instance among our datasets. Even when the Profile ILP returns a near-optimal solution, the average global representation of the most disadvantaged group is 0.12. We find that this happens due to the points from a single group being distributed evenly across all clusters, which results in a hard instance of Fair Allocation (see, e.g., Figure 3); this can be addressed by increasing the k/t ratio, as $k = 10t$ results in a 1.6× larger global representation.

Lastly, we study how optimizing one representation objective with the Profile ILP affects performance on the other for varying values of k/t; here, our goal is to understand how tightly connected our two notions of representation are. In Figure 4 for k-Means, we observe that as k/t increases, the points move predominantly to the right for all datasets, i.e., optimizing global representation while increasing k/t also improves the local representation. On the other hand, the points stay largely flat along the y-axis, meaning that the global representation does not improve when optimizing the local representation under increasing k/t. Next, we observe that the obtained representations are substantially lower than when optimizing for the respective objective directly. For instance, for `MovieLens` the global representation is less than 0.2 for all values of k/t when optimizing for $\text{minrep}_{\text{local}}$ (see Figure 4), whereas it is already 0.3 when directly optimizing for $\text{minrep}_{\text{global}}$ even for $k/t = 2$ and 0.36 for $k/t = 10$ (see Table 1); nonetheless, the global representations yielded by local optimization are no worse than those of the baselines. The situation for other datasets and for local representation is similar; further, we observe similar trends for k-Supplier (see Appendix §B.2). To answer **RQ2**, we conclude that the loss in algorithmic fairness for one objective due to optimization of the other is substantial and in practice it is important to carefully decide which representation to optimize.

5.2 Scalability

Next, our goal is to evaluate the algorithms' running times and solution qualities in relation to the dataset size. We generate synthetic data by independently sampling $n \in \{10^3, 10^4, 10^5, 10^6, 10^7\}$ points uniformly from $[0, 1)^{10}$. To determine group membership, we draw the group sizes uniformly at random, and then assign points to groups randomly by shuffling the data. We vary $t \in \{2^i : i \in [5]\}$ and $k \in \{2t, 4t, 6t, 8t, 10t\}$, while restricting the number of facilities n_f to $0.2n$ for k-Supplier, and n for k-Center.[4] To maintain a tractable configuration grid, we only consider the equitable variant of the requirement vector $(\lfloor k/t \rfloor)_{j \in [t]}$ for Fair k-Clustering; this is because when we study the solution qualities, the 3-approximation for Fair k-Supplier [18] yields higher max-min representation under equitable requirements. Further, for the Assignment LP we use $\ell = 64$ randomized rounding trials; in preliminary experiments

[4] We exclude k-Means from the scalability experiments because after identifying the candidate groups, the allocation stage is identical for both clustering objectives.

Table 1: Minimum global group representation $\text{minrep}_{\text{global}}(C, S')$ in the clusterings returned by the baselines and our algorithms, for the k-MEANS objective. Results are reported as mean ± standard deviation **over ten independent runs.**

				$k = 2t$				$k = 10t$			
Dataset	n	m	t	k-means++	Heuristic DIV k-MEANS	Profile ILP	Assignment LP	k-means++	Heuristic DIV k-MEANS	Profile ILP	Assignment LP
Heart	303	11	4	0.13 ± 0.11	0.19 ± 0.06	**0.32** ± 0.02	0.27 ± 0.04	0.27 ± 0.09	0.29 ± 0.04	**0.50** ± 0.02	0.48 ± 0.03
Student	649	37	4	0.03 ± 0.05	0.22 ± 0.02	**0.28** ± 0.03	0.24 ± 0.05	0.13 ± 0.04	0.32 ± 0.02	**0.39** ± 0.01	0.38 ± 0.02
MovieLens	6,040	701	4	0.09 ± 0.07	0.13 ± 0.03	**0.30** ± 0.03	0.27 ± 0.05	0.15 ± 0.05	0.18 ± 0.02	**0.36** ± 0.01	0.35 ± 0.02
CreditCard	30,000	20	8	0.00 ± 0.01	0.04 ± 0.02	**0.12** ± 0.01	0.06 ± 0.03	0.02 ± 0.02	0.09 ± 0.02	**0.19** ± 0.00	0.15 ± 0.02
Bank	45,211	39	6	0.00 ± 0.00	0.12 ± 0.03	**0.19** ± 0.01	0.16 ± 0.02	0.02 ± 0.02	0.14 ± 0.02	**0.24** ± 0.00	0.22 ± 0.01
Adult	48,842	53	16	0.00 ± 0.00	0.03 ± 0.01	**0.12** ± 0.00	0.08 ± 0.01	0.00 ± 0.01	0.06 ± 0.01	**0.19** ± 0.00	0.17 ± 0.01

Table 2: Minimum local group representation $\text{minrep}_{\text{local}}(C, S')$ in the clusterings returned by the baselines and our algorithms, for the k-MEANS objective. Results are reported as mean ± standard deviation **over ten independent runs.**

				$k = 2t$				$k = 10t$			
Dataset	n	m	t	k-means++	Heuristic DIV k-MEANS	Profile ILP	Assignment LP	k-means++	Heuristic DIV k-MEANS	Profile ILP	Assignment LP
Heart	303	11	4	0.20 ± 0.16	0.36 ± 0.13	**0.53** ± 0.05	0.50 ± 0.06	2.16 ± 0.58	3.29 ± 0.25	**4.06** ± 0.53	4.03 ± 0.52
Student	649	37	4	0.08 ± 0.16	0.29 ± 0.02	**0.48** ± 0.05	0.47 ± 0.05	1.33 ± 0.71	2.27 ± 0.24	**3.51** ± 0.34	3.49 ± 0.32
MovieLens	6,040	701	4	0.18 ± 0.10	0.27 ± 0.03	**0.48** ± 0.05	0.44 ± 0.03	0.96 ± 0.32	1.49 ± 0.05	**3.00** ± 0.17	2.98 ± 0.18
CreditCard	30,000	20	8	0.01 ± 0.02	0.06 ± 0.01	**0.21** ± 0.01	0.20 ± 0.01	0.15 ± 0.10	0.32 ± 0.02	**1.53** ± 0.10	1.49 ± 0.11
Bank	45,211	39	6	0.00 ± 0.00	0.07 ± 0.01	**0.17** ± 0.00	0.15 ± 0.03	0.08 ± 0.04	0.39 ± 0.05	**1.15** ± 0.02	1.12 ± 0.03
Adult	48,842	53	16	0.00 ± 0.00	0.01 ± 0.00	**0.07** ± 0.01	0.00 ± 0.00	0.02 ± 0.02	0.13 ± 0.01	**0.60** ± 0.05	0.54 ± 0.05

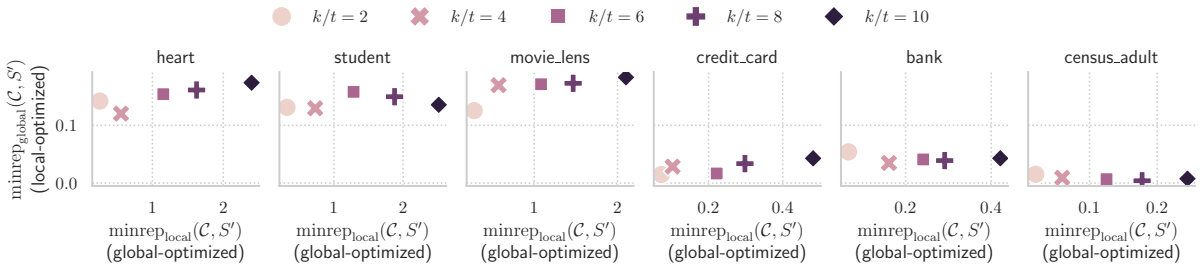


Figure 4: Scatter plot that shows how optimizing $\text{minrep}_{\text{global}}$ affects $\text{minrep}_{\text{local}}$, and vice-versa, under the k-MEANS objective. As k/t increases, optimizing $\text{minrep}_{\text{global}}$ yields better gains for $\text{minrep}_{\text{local}}$ (x axis) than the other direction (y axis).

we found that more than 64 rounding trials offer limited gains in the objective (see Appendix §B.1). Any configuration that exceeds a runtime of 30 minutes is timed out.

Figure 5 compares the runtime of our algorithms (in seconds) against 3APX FAIR k-SUPPLIER [18] as n grows, again for the two extremal cases $k/t \in \{2, 10\}$. The shaded areas indicate 95% confidence intervals for the mean runtime. We note that the Assignment LP is consistently the fastest method. This is expected since it solves an LP of size $O(kt)$, whereas 3APX FAIR k-SUPPLIER performs binary search over $O(kn)$ radii to find a maximum matching.

The Profile ILP exhibits different behavior depending on the representation function. On local representation, it scales comparably to the Assignment LP, outperforming the baseline at all input sizes. Although it is slower on the global representation at intermediate input sizes, its runtime saturates beyond $n = 10^6$ while the baseline's runtime incurs the largest increase between $n = 10^6$ and $n = 10^7$.

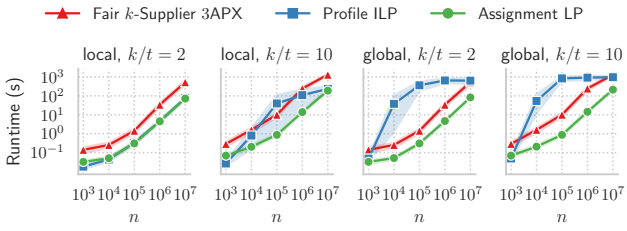


Figure 5: Scalability of our algorithms w.r.t FAIR k-SUPPLIER 3APX [18] on minrep for $r_{\text{local}}, r_{\text{global}}$ and $k/t \in \{2, 10\}$.

Next, we examine the solution quality of all three algorithms for $k \in \{2t, 10t\}$ as n grows, by comparing it with the optimal (fractional) Assignment LP solution OPT_{LP} (see Figure 6).

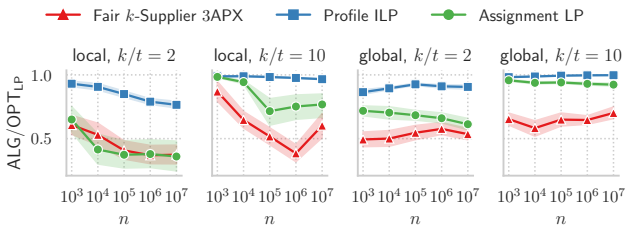


Figure 6: Closeness to optimality (ALG/OPT$_{LP}$) where ALG denotes the solution value (minrep for r_{local}, r_{global}) of the allocation returned by each algorithm for $k/t \in \{2, 10\}$.

We find that the Profile ILP consistently returns an allocation with the closest solution value to OPT$_{LP}$, with near-optimal allocations for higher values of k/t. With the exception of minrep$_{local}$ for $k = 2t$, the Assignment LP always outperforms 3APX Fair k-Supplier. For larger values of k/t, it demonstrates comparable performance to the Profile ILP on minrep$_{global}$.

To answer **RQ3**, we find that our algorithms are highly scalable, solving instances with 10 million points in less than 20 minutes. In practice, the Profile ILP is the best choice since it achieves the best solution qualities. If running times are a constraint then one may consider using the Assignment LP, which is faster and, even though it returns worse solutions than the Profile ILP, its solutions are substantially more representative than those of the baselines.

6 Related Work

The literature on fair clustering is extensive and provides several definitions that depend on the underlying application; see Chhabra et al. [13] for a more detailed review. Below, we discuss prior work on data summarization in detail, and also briefly cover related algorithmic fairness definitions in other contexts.

Kleindessner et al. [29] first introduced the Fair k-Clustering problem for the k-Center objective, and presented a $(3 \cdot 2^{t-1})$-approximation in time $O(nkt^2 + km^4)$. Jones et al. [25] improved this factor to 3 with a linear time algorithm based on maximum-matching, which was generalized to a 5-approximation for Fair k-Supplier in time $O(kn^2 + k^2\sqrt{k})$ by Chen et al. [12]. More recently, Gadekar et al. [18] presented a 3-approximation for Fair k-Supplier in time $O((kn + k^2\sqrt{k}) \log n \log k)$ that matches the theoretical lower bound. They also supplement their results with an algorithm for intersecting groups with an additional factor of $(2^{tk}k)$ in the runtime. Similarly, Thejaswi et al. [35, 36] provide FPT algorithms with approximation ratios $1 + 2/e + \varepsilon$ and $1 + 8/e + \varepsilon$ under the k-Median and k-Means objectives respectively. They further show that Fair k-Clustering cannot be approximated to any multiplicative factor in polynomial time under these clustering objectives.

Our work also bears some similarity to notions for proportionally fair clustering [4, 11, 14], and fair clustering with equitable group representations [1]. The former ensures that any $\lfloor n/k \rfloor$ points can form their own cluster if there is a closer center, whereas the latter seeks a clustering whose cost is not disproportionately borne by some group. These lines of inquiry are however, more concerned with relative distances of points or groups to their centers, with applications to facility location (e.g., polling locations for voting).

7 Discussion and Conclusion

We introduced the α-Representative Fair k-Clustering problem, which seeks a clustering that allocates at least α representation to each group, and generalizes the Fair k-Clustering problem. We further provided two algorithms for solving this problem, which allow for bounding the increase in the clustering cost while ensuring near-optimal representation of the groups. In our experiments, we showed that our algorithms yield group representations that are at least twice as high as those of state-of-the-art baselines. Next, we discuss the implications of our work for the web, followed by its limitations and directions for future work.

Implications for the Web. Kleindessner et al. [29] first motivated Fair k-Clustering with the example that a Google Image search for "CEO" returned disproportionately more images of men than women [27]. Count-based constraints ($\vec{\beta}$) in this definition aim to correct this imbalance.

However, we show that count-based algorithmic fairness criteria are limited in their ability to capture representation. We argue that this extends not only to digital participation platforms, but also to search and recommendation systems, which have been studied extensively in the context of exposure inequality [15, 17, 34, 38, 39].

Cherry-picking representatives to satisfy count constraints in these applications can similarly create an illusion of fairness. For instance, a music streaming platform may meet a quota for a less popular genre (say shoegaze) by recommending a shoegaze song that sounds much closer to pop. Our representation functions can expose such a choice, since the cluster of the recommended shoegaze song would consist mostly of pop songs. More broadly, our proposed notion of α-Representative Fair k-Clustering can serve as an auditing tool for *tokenism* in information access systems, and our algorithms, as mechanisms for provably repairing it.

Limitations and Future Work. While our methods allow us to audit and repair tokenism in data summaries, we caution that they can also lead to *stereotyping*, i.e., reinforcing over-generalized beliefs about groups [19]. For instance, consider an image search for "healthcare professionals". Suppose the cluster with doctors consists predominantly of men, and the cluster with nurses consists mostly of women. Selecting a nurse who is a woman, and a doctor who is a man in the data summary can reinforce occupational gender stereotypes while achieving high representation under our objective. To prevent the same, one may set the representation function in α-Representative Fair k-Clustering accordingly; for instance, by defining each clusters' utility for a group based on gender-occupation pairs that challenge existing stereotypes. However, we leave this idea for future work.

Lastly, as we show in the proof of Theorem 8, the Fair Allocation problem in our algorithmic framework directly reduces to the problem of fairly allocating indivisible goods (clusters) to agents (groups) under the max-min objective [5, 7]. Hence, another interesting line of future work is to study applications where other objectives from social choice theory (e.g., envy freeness, equity) may be useful and tractable when combined with clustering.

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A Omitted Proofs From the Main Text

We present proofs that we had to omit from the main text due to space constraints.

A.1 Proof of Lemma 3

Consider a Fair k-Clustering instance with requirement vector $\vec{\beta} = (\beta_i)_{i\in[t]}$. Now, setting $\mathrm{r}(C_i, G_j) = 1/\beta_j$ for all $i \in [k]$ and $j \in [t]$ reduces minrep(C, S) in Definition 2 to

$$\sum_{i\in[k]} \mathbf{1}(s_i \in G_j)\,\mathrm{r}(C_i, G_j) = \sum_{i\in[k]} \mathbf{1}(s_i \in G_j)/\beta_j = |S \cap G_j|/\beta_j.$$

For $\alpha = 1$, this constraint becomes $|S \cap G_j| \geq \beta_j$ for all $j \in [t]$, which is equivalent to the constraint in Definition 1. Therefore, solving the α-Representative Fair k-Clustering problem for $\mathrm{r}(C_i, G_j) = 1/\beta_j$ and $\alpha = 1$, one can obtain a solution to the Fair k-Clustering problem.

A.2 Proof of Lemma 4

We use the triangle inequality to bound the distance between any point $x \in C_i$ and its new center s_i' as follows:

$$d(x, s_i') \leq d(x, s_i) + d(s_i, s_i') \leq \Gamma + \lambda.$$

Since this holds for all $C_i \in C$, we have that

$$\mathrm{cost}_\infty(C, S') = \max_{i\in[k]} \max_{x\in C_i} d(x, s_i') \leq \Gamma + \lambda.$$

A.3 Proof of Lemma 5

For any cluster $C_i \in C$ with $s_i = \frac{1}{|C_i|}\sum_{x\in C_i} x$ [5], we can bound its cost with respect to the new center s_i' using Lemma 2.1 from Arthur and Vassilvitskii [2]:

$$\sum_{x\in C_i} d(x,s_i')^2 = \sum_{x\in C_i} d(x,s_i)^2 + |C_i|\cdot d(s_i,s_i')^2 \le (1+\lambda)\Gamma_i.$$

Since this holds for all $C_i \in C$, we have that

$$\text{cost}_2(C,S') = \sum_{i\in[k]}\sum_{x\in C_i} d(x,s_i)^2 \le (1+\lambda)\sum_{i\in[k]}\Gamma_i = (1+\lambda)\Gamma.$$

A.4 Proof of Theorem 8

In our reduction, we make use of the Max-Min Fairness problem [7]. An instance I consists of t agents and a set of k indivisible goods C, and a utility function $u : 2^C \times [t] \to \mathbb{R}_{\ge 0}$. The objective is to compute a partition $C_1,\ldots,C_t$ of the goods that maximizes $\min_{j\in[t]}\sum_{C\in C_j} u(C,j)$.

Normalizing the utilities by $M = \max_{C\in C, j\in[t]} u(C,j)$, we can obtain an instance of Fair Allocation where goods correspond to clusters, agents to groups, and the representation function is given by $r(C,j) = u(C,j)/M$.

Now, suppose such an algorithm exists. Running it with accuracy parameter ϵ/M and rescaling yields a solution of value at least $\text{OPT}-\epsilon k$ to the Max-Min Fairness problem in time $\text{poly}(M/\epsilon,t,k)$. Bezáková and Dani [7] give a polynomial-time reduction from 3-Dimensional Matching to Max-Min Fairness, such that there exists a 3D-matching iff $\text{OPT}\ge 2$, and otherwise, $\text{OPT}\le 1$. We set $\epsilon k = 1/4$ to obtain an additive error of $\pm 1/4$ for OPT. Since their reduction uses utilities in $\{0,1,2\}$, we have $M=2$, and the running time simplifies to $\text{poly}(1/\epsilon,t,k)$. If a 3D-matching exists, it returns an allocation of value at least $7/4$. If it does not exist, the solution value is at most 1. Thus, comparing the returned value with $3/2$ can decide the 3-Dimensional Matching problem in polynomial time, implying P = NP.

B Additional Experimental Results

We present additional experimental results.

B.1 Rounding Trials in the Assignment LP

To study how the number of independent randomized rounding trials (ℓ) affect the rounded Assignment LP solution (ALG_ℓ), we vary $\ell \in \{2^i : i \in [10]\}$. Figures 7 and 8 report $\text{ALG}_\ell/\text{OPT}_{\text{LP}}$ for the k-Supplier and k-Means objectives respectively, where OPT_{LP} denotes the optimal (fractional) solution to the Assignment LP. The shaded area indicates the standard deviation for different configurations ($k/t \in \{2,4,6,8,10\}$, $n_f/n \in \{0.2,0.4,0.6,0.8,1\}$) across ten independent runs.

We find that for most datasets, the ratio $\text{ALG}_\ell/\text{OPT}_{\text{LP}}$ saturates by $\ell = 64$, and larger values of ℓ do not offer substantive gains for both $\text{minrep}_{\text{global}}$ and $\text{minrep}_{\text{local}}$.

[5] If $s_i \ne \frac{1}{|C_i|}\sum_{x\in C_i} x$, we simply run the update step from Lloyd's algorithm to obtain a clustering of cost at most Γ and $s_i = \frac{1}{|C_i|}\sum_{x\in C_i} x$ for all $i \in [k]$.

B.2 Algorithmic Fairness for k-Supplier

In this section, we present the algorithmic fairness evaluation for all algorithms under the k-Supplier objective. Tables 3 and 4 report $\text{minrep}(C,S')$ for the global and local representation functions respectively. Again, we note that the Profile ILP outperforms all algorithms on all datasets, as well as both representation objectives. Second, while 3APX Fair k-Supplier [18] offers notable improvement over the k-Supplier 3-approximation, its performance still remains inferior to the Assignment LP, with the exception of $k = 2t$ on `Adult` for the local objective. Third, the performance gap between the Assignment LP, and the Profile ILP is generally smaller for the local objective, and for larger values of k/t.

Next, we closely examine `CreditCard` ($k = 2t$) which is the hardest instance among our datasets. In the solution returned by the Profile ILP, the average global representation of the most disadvantaged group is 0.015 for k-Supplier. We find that this happens due to the majority of data points being concentrated in a single cluster, which results in a hard instance of Fair Allocation (see for e.g., Figure 9); this can be addressed by improving the k/t ratio, as $k = 10t$ improves $\text{minrep}(C,S')$ by over ten percent.



Figure 9: Illustration of a hard Fair Allocation instance under global representation, where each ▪ denotes 250 points, each ● denotes 1 point, and the colors indicate groups. Regardless of how the clusters may be allocated, no more than 0.001 fraction of one group will be represented by a same-group center, implying that OPT ≤ 0.001.

Finally, we study how optimizing global representation affects the overall local representation, and vice-versa (see Figure 10). As k/t increases, optimizing global representation substantially improves local representation on all datasets. On the other hand, global representation under local-optimization only improves on `Heart`, `Student`, and to some extent, `MovieLens`.

B.3 Clustering Cost and Feasibility of Fairness

In this section, we study how increasing the clustering cost for k-Means improves the feasibility of recovering a fair solution i.e., a clustering (C,S') with $\text{minrep}(C,S') > 0$, and its objective value $\text{minrep}(C,S')$. Figures 11 and 12 report the results for global and local representation respectively.

We note that recovering a fair clustering is feasible at lower clustering costs for smaller cluster to group ratios (k/t). This is expected, since a smaller k-partition of points makes it more likely to find points from different groups that lie close to the center. However, once a feasible fair clustering can be recovered at some λ, increasing it does not always lead to higher max-min representation.

B.4 Algorithmic Fairness for all Configurations

In this section, we present the algorithmic fairness evaluation for all configurations based on the cluster to group ratio k/t, number of facilities n_f, and the requirement vector $\vec{\beta}$. Due to space constraints, we use the following abbreviations for the algorithms:

(1) FFT: Farthest First Traversal (3APX k-Supplier) [24].
(2) FKSP: 3APX Fair k-Supplier [18].
(3) PILP: Profile ILP.
(4) ALP: Assignment LP.
(5) KM++: k-means++ [2].
(6) HDKM: Heuristic Div k-Means [35].

Tables 5 and 6 report the minimum group representation under the k-Supplier objective, for r_{global} and r_{local} respectively. Table 7 reports the minimum group representation under the k-Means objective. Note that the results for both the Profile ILP and Assignment LP may differ between the equitable and proportional variants of the requirement vector for Fair k-Supplier. This is because both instances contain different number of facilities per group, which results in different inputs for Fair Allocation.

Overall, we find that achieving higher minrep(C, S') becomes easier as k/t increases. In fact, on every dataset, and for both r_{global} and r_{local}, the average minrep(C, S') computed by the Profile ILP is lowest for $k = 2t$, and highest for $k = 10t$, with the intermediate values lying in between. Among the ratios we consider, $k = 2t$ and $k = 10t$ form the easiest and hardest cases for algorithmic fairness respectively. Hence, we report them in the main text of our paper.

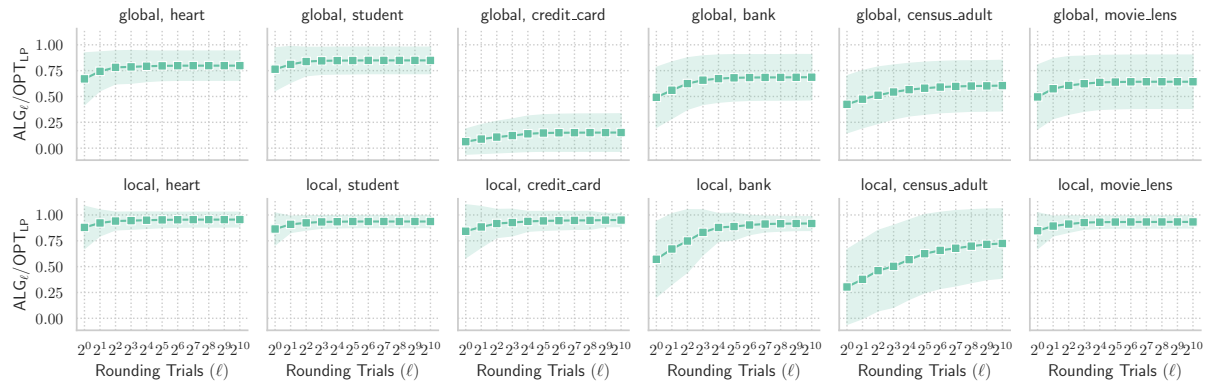


Figure 7: Closeness to the optimal (fractional) Assignment LP solution ($\mathrm{minrep}_{\mathrm{global}}$ and $\mathrm{minrep}_{\mathrm{local}}$) for varying number of independent randomized rounding trials (ℓ) under the k-Supplier clustering objective.



Figure 8: Closeness to the optimal (fractional) Assignment LP solution ($\mathrm{minrep}_{\mathrm{global}}$ and $\mathrm{minrep}_{\mathrm{local}}$) for varying number of independent randomized rounding trials (ℓ) under the k-Means clustering objective.

Table 3: Minimum global group representation $\mathrm{minrep}_{\mathrm{global}}(C, S')$ in the clusterings returned by the baselines and our algorithms, for the k-Supplier objective ($n_f = 0.2n$). Results are reported as mean ± standard deviation over ten independent runs.

				$k = 2t$				$k = 10t$			
Dataset	n	m	t	3APX k-Supplier	3APX Fair k-Supplier	Profile ILP	Assignment LP	3APX k-Supplier	3APX Fair k-Supplier	Profile ILP	Assignment LP
Heart	303	11	4	0.02 ± 0.04	0.10 ± 0.06	**0.26** ± 0.05	0.22 ± 0.06	0.24 ± 0.11	0.17 ± 0.03	**0.39** ± 0.03	0.38 ± 0.03
Student	649	37	4	0.04 ± 0.05	0.12 ± 0.03	**0.27** ± 0.02	0.23 ± 0.04	0.19 ± 0.05	0.20 ± 0.04	**0.38** ± 0.02	0.36 ± 0.02
MovieLens	6,040	701	4	0.00 ± 0.01	0.07 ± 0.05	**0.22** ± 0.05	0.17 ± 0.06	0.13 ± 0.08	0.18 ± 0.03	**0.37** ± 0.01	0.34 ± 0.02
CreditCard	30,000	20	8	0.00 ± 0.00	0.00 ± 0.00	**0.01** ± 0.01	0.00 ± 0.00	0.01 ± 0.01	0.02 ± 0.01	**0.13** ± 0.02	0.07 ± 0.03
Bank	45,211	39	6	0.00 ± 0.00	0.04 ± 0.02	**0.16** ± 0.02	0.12 ± 0.03	0.02 ± 0.02	0.09 ± 0.02	**0.23** ± 0.00	0.20 ± 0.02
Adult	48,842	53	16	0.00 ± 0.00	0.01 ± 0.01	**0.10** ± 0.01	0.04 ± 0.03	0.01 ± 0.01	0.02 ± 0.01	**0.18** ± 0.00	0.15 ± 0.01

Table 4: Minimum local group representation $\text{minrep}_{\text{local}}(C, S')$ in the clusterings returned by the baselines and our algorithms, for the k-Supplier objective ($n_f = 0.2n$). Results are reported as mean ± standard deviation **over ten independent runs.**

Dataset	n	m	t	$k = 2t$				$k = 10t$			
				3APX k-Supplier	3APX Fair k-Supplier	Profile ILP	Assignment LP	3APX k-Supplier	3APX Fair k-Supplier	Profile ILP	Assignment LP
Heart	303	11	4	0.06 ± 0.10	0.26 ± 0.15	**0.48** ± 0.10	0.46 ± 0.09	1.06 ± 0.46	0.94 ± 0.28	**1.92** ± 0.37	**1.92** ± 0.37
Student	649	37	4	0.09 ± 0.12	0.20 ± 0.06	**0.46** ± 0.04	0.44 ± 0.04	1.61 ± 0.48	1.36 ± 0.32	**2.81** ± 0.56	2.80 ± 0.55
MovieLens	6,040	701	4	0.01 ± 0.04	0.23 ± 0.07	**0.51** ± 0.04	0.48 ± 0.06	1.01 ± 0.31	1.29 ± 0.26	**2.98** ± 0.20	2.94 ± 0.19
CreditCard	30,000	20	8	0.00 ± 0.01	0.06 ± 0.04	**0.22** ± 0.05	0.19 ± 0.06	0.34 ± 0.27	0.29 ± 0.21	**1.46** ± 0.39	1.44 ± 0.38
Bank	45,211	39	6	0.00 ± 0.00	0.06 ± 0.02	**0.18** ± 0.01	0.16 ± 0.03	0.05 ± 0.04	0.31 ± 0.07	**1.14** ± 0.04	1.12 ± 0.04
Adult	48,842	53	16	0.00 ± 0.00	0.01 ± 0.00	**0.07** ± 0.01	0.00 ± 0.02	0.01 ± 0.02	0.06 ± 0.03	**0.58** ± 0.05	0.54 ± 0.06

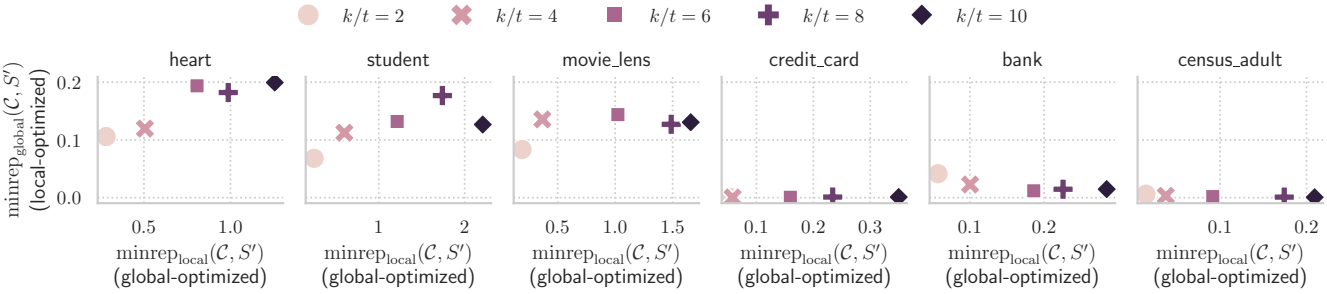


Figure 10: Scatter plot that shows how optimizing $\text{minrep}_{\text{global}}$ affects $\text{minrep}_{\text{local}}$, and vice-versa, under the k-Supplier objective. As k/t increases, optimizing $\text{minrep}_{\text{global}}$ yields better gains for $\text{minrep}_{\text{local}}$ (x axis) than the other direction (y axis).

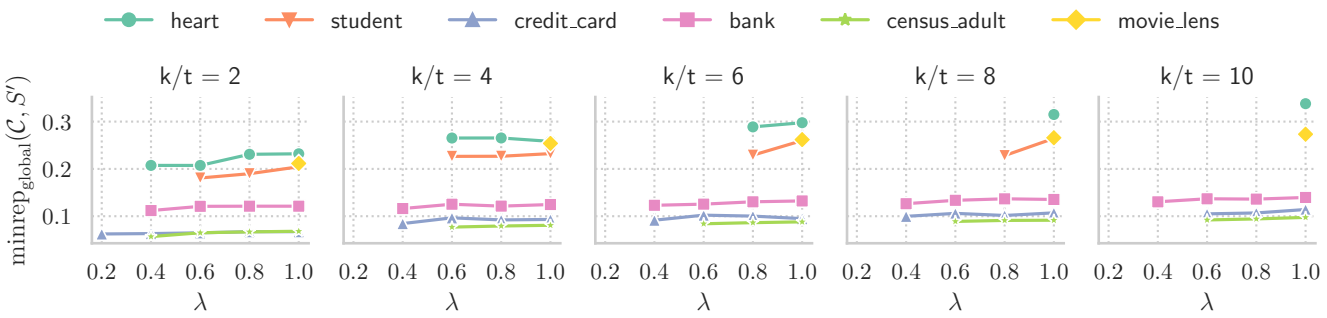


Figure 11: Effect of increasing the k-Means clustering cost by $(1 + \lambda)$ on feasibility of fairness and global representation.

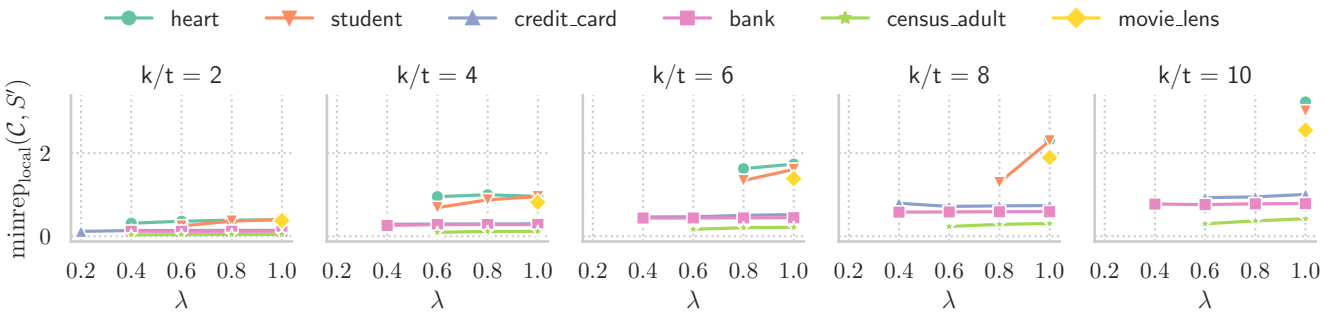


Figure 12: Effect of increasing the k-Means clustering cost by $(1 + \lambda)$ on feasibility of fairness and local representation.

Table 5: Minimum global group representation $\text{minrep}_{\text{global}}(C, S')$ **in the clusterings returned by the baselines and our algorithms, for the** k**-Supplier objective (all values of** $n_f, k/t$**). Results are reported as** mean ± standard deviation **over ten independent runs.**

			Equitable requirement vector ($\vec{\beta}$)				Proportional requirement vector ($\vec{\beta}$)			
Dataset	n_f/n	k/t	FFT	FKSP	PILP	ALP	FFT	FKSP	PILP	ALP
Heart	0.2	2	0.02 ± 0.04	0.10 ± 0.06	**0.26** ± 0.05	0.22 ± 0.06	0.04 ± 0.07	0.08 ± 0.05	**0.28** ± 0.06	0.25 ± 0.08
		4	0.13 ± 0.10	0.12 ± 0.04	**0.34** ± 0.03	0.31 ± 0.05	0.10 ± 0.10	0.13 ± 0.08	**0.33** ± 0.04	0.30 ± 0.03
		6	0.20 ± 0.11	0.16 ± 0.05	**0.39** ± 0.02	0.36 ± 0.04	0.17 ± 0.13	0.09 ± 0.06	**0.37** ± 0.04	0.36 ± 0.06
		8	0.21 ± 0.10	0.12 ± 0.05	**0.39** ± 0.03	0.36 ± 0.04	0.20 ± 0.09	0.17 ± 0.06	**0.38** ± 0.05	0.35 ± 0.05
		10	0.24 ± 0.11	0.17 ± 0.03	**0.39** ± 0.03	0.38 ± 0.03	0.25 ± 0.09	0.12 ± 0.07	**0.38** ± 0.03	0.36 ± 0.03
	0.4	2	0.06 ± 0.06	0.15 ± 0.03	**0.26** ± 0.05	0.21 ± 0.06	0.05 ± 0.07	0.10 ± 0.06	**0.28** ± 0.04	0.22 ± 0.07
		4	0.13 ± 0.15	0.12 ± 0.07	**0.38** ± 0.03	0.34 ± 0.05	0.09 ± 0.11	0.12 ± 0.07	**0.38** ± 0.03	0.35 ± 0.04
		6	0.17 ± 0.15	0.16 ± 0.04	**0.43** ± 0.02	0.37 ± 0.05	0.13 ± 0.11	0.12 ± 0.05	**0.41** ± 0.02	0.35 ± 0.04
		8	0.25 ± 0.08	0.15 ± 0.06	**0.43** ± 0.01	0.40 ± 0.03	0.19 ± 0.11	0.13 ± 0.06	**0.44** ± 0.03	0.40 ± 0.04
		10	0.25 ± 0.14	0.17 ± 0.03	**0.46** ± 0.02	0.43 ± 0.03	0.21 ± 0.13	0.14 ± 0.07	**0.46** ± 0.03	0.43 ± 0.05
	0.6	2	0.05 ± 0.06	0.15 ± 0.05	**0.26** ± 0.03	0.22 ± 0.06	0.04 ± 0.06	0.07 ± 0.06	**0.26** ± 0.03	0.20 ± 0.04
		4	0.08 ± 0.08	0.14 ± 0.04	**0.39** ± 0.03	0.34 ± 0.05	0.06 ± 0.07	0.17 ± 0.09	**0.39** ± 0.02	0.35 ± 0.04
		6	0.09 ± 0.08	0.14 ± 0.02	**0.43** ± 0.02	0.40 ± 0.04	0.11 ± 0.12	0.16 ± 0.08	**0.44** ± 0.02	0.40 ± 0.05
		8	0.13 ± 0.12	0.17 ± 0.05	**0.46** ± 0.02	0.42 ± 0.03	0.15 ± 0.14	0.16 ± 0.06	**0.46** ± 0.02	0.44 ± 0.03
		10	0.23 ± 0.11	0.15 ± 0.05	**0.48** ± 0.01	0.44 ± 0.03	0.15 ± 0.15	0.14 ± 0.04	**0.48** ± 0.02	0.45 ± 0.02
	0.8	2	0.04 ± 0.06	0.14 ± 0.07	**0.27** ± 0.04	0.21 ± 0.05	0.03 ± 0.04	0.07 ± 0.06	**0.26** ± 0.03	0.21 ± 0.05
		4	0.06 ± 0.07	0.13 ± 0.05	**0.39** ± 0.02	0.34 ± 0.06	0.05 ± 0.06	0.17 ± 0.08	**0.38** ± 0.03	0.34 ± 0.05
		6	0.10 ± 0.10	0.15 ± 0.07	**0.43** ± 0.02	0.38 ± 0.05	0.07 ± 0.09	0.14 ± 0.08	**0.43** ± 0.02	0.40 ± 0.04
		8	0.11 ± 0.13	0.17 ± 0.04	**0.46** ± 0.02	0.41 ± 0.03	0.14 ± 0.12	0.14 ± 0.05	**0.46** ± 0.02	0.42 ± 0.03
		10	0.13 ± 0.12	0.18 ± 0.04	**0.49** ± 0.01	0.45 ± 0.03	0.10 ± 0.07	0.14 ± 0.08	**0.49** ± 0.01	0.46 ± 0.02
	1.0	2	0.04 ± 0.05	0.14 ± 0.06	**0.27** ± 0.04	0.23 ± 0.06	0.04 ± 0.05	0.06 ± 0.06	**0.27** ± 0.04	0.23 ± 0.06
		4	0.05 ± 0.06	0.14 ± 0.04	**0.38** ± 0.03	0.34 ± 0.05	0.05 ± 0.06	0.15 ± 0.09	**0.38** ± 0.03	0.34 ± 0.05
		6	0.08 ± 0.08	0.13 ± 0.05	**0.43** ± 0.02	0.39 ± 0.04	0.08 ± 0.08	0.13 ± 0.05	**0.43** ± 0.02	0.39 ± 0.04
		8	0.09 ± 0.09	0.19 ± 0.06	**0.46** ± 0.02	0.42 ± 0.03	0.09 ± 0.09	0.15 ± 0.09	**0.46** ± 0.02	0.42 ± 0.03
		10	0.11 ± 0.11	0.22 ± 0.05	**0.49** ± 0.01	0.45 ± 0.02	0.11 ± 0.11	0.17 ± 0.08	**0.49** ± 0.01	0.45 ± 0.02
Student	0.2	2	0.04 ± 0.05	0.12 ± 0.03	**0.27** ± 0.02	0.23 ± 0.04	0.04 ± 0.06	0.09 ± 0.04	**0.26** ± 0.02	0.21 ± 0.03
		4	0.09 ± 0.06	0.15 ± 0.05	**0.33** ± 0.02	0.29 ± 0.04	0.10 ± 0.05	0.08 ± 0.03	**0.33** ± 0.02	0.31 ± 0.02
		6	0.12 ± 0.08	0.18 ± 0.05	**0.35** ± 0.02	0.34 ± 0.02	0.10 ± 0.06	0.09 ± 0.05	**0.35** ± 0.02	0.32 ± 0.03
		8	0.17 ± 0.09	0.18 ± 0.03	**0.37** ± 0.01	0.35 ± 0.02	0.16 ± 0.06	0.12 ± 0.04	**0.36** ± 0.02	0.35 ± 0.03
		10	0.19 ± 0.05	0.20 ± 0.04	**0.38** ± 0.02	0.36 ± 0.02	0.16 ± 0.07	0.09 ± 0.03	**0.37** ± 0.02	0.36 ± 0.02
	0.4	2	0.02 ± 0.06	0.10 ± 0.05	**0.27** ± 0.04	0.23 ± 0.06	0.06 ± 0.09	0.07 ± 0.03	**0.26** ± 0.03	0.21 ± 0.04
		4	0.09 ± 0.06	0.15 ± 0.05	**0.34** ± 0.02	0.29 ± 0.04	0.08 ± 0.07	0.09 ± 0.06	**0.33** ± 0.01	0.29 ± 0.03
		6	0.13 ± 0.05	0.17 ± 0.05	**0.36** ± 0.01	0.34 ± 0.02	0.12 ± 0.04	0.10 ± 0.05	**0.36** ± 0.01	0.33 ± 0.02
		8	0.16 ± 0.05	0.15 ± 0.04	**0.38** ± 0.01	0.35 ± 0.02	0.15 ± 0.06	0.13 ± 0.04	**0.38** ± 0.01	0.35 ± 0.03
		10	0.17 ± 0.05	0.18 ± 0.03	**0.39** ± 0.01	0.37 ± 0.02	0.19 ± 0.05	0.11 ± 0.03	**0.40** ± 0.01	0.38 ± 0.02
	0.6	2	0.05 ± 0.08	0.12 ± 0.04	**0.26** ± 0.03	0.21 ± 0.04	0.07 ± 0.07	0.09 ± 0.06	**0.27** ± 0.03	0.23 ± 0.05
		4	0.06 ± 0.07	0.15 ± 0.03	**0.33** ± 0.01	0.28 ± 0.02	0.07 ± 0.05	0.08 ± 0.05	**0.33** ± 0.02	0.28 ± 0.03
		6	0.10 ± 0.04	0.16 ± 0.03	**0.37** ± 0.02	0.34 ± 0.04	0.13 ± 0.08	0.11 ± 0.05	**0.36** ± 0.02	0.32 ± 0.04
		8	0.18 ± 0.07	0.16 ± 0.03	**0.38** ± 0.02	0.35 ± 0.03	0.12 ± 0.05	0.14 ± 0.03	**0.38** ± 0.01	0.34 ± 0.02
		10	0.19 ± 0.06	0.19 ± 0.02	**0.40** ± 0.01	0.38 ± 0.02	0.17 ± 0.07	0.10 ± 0.03	**0.40** ± 0.01	0.38 ± 0.01
	0.8	2	0.04 ± 0.04	0.12 ± 0.05	**0.26** ± 0.04	0.21 ± 0.06	0.05 ± 0.04	0.08 ± 0.04	**0.26** ± 0.03	0.20 ± 0.06
		4	0.11 ± 0.06	0.15 ± 0.04	**0.34** ± 0.01	0.30 ± 0.03	0.09 ± 0.08	0.11 ± 0.05	**0.33** ± 0.02	0.29 ± 0.03
		6	0.13 ± 0.08	0.14 ± 0.05	**0.36** ± 0.02	0.31 ± 0.04	0.14 ± 0.07	0.11 ± 0.05	**0.36** ± 0.01	0.33 ± 0.03
		8	0.17 ± 0.08	0.17 ± 0.03	**0.39** ± 0.01	0.36 ± 0.03	0.17 ± 0.07	0.13 ± 0.03	**0.39** ± 0.02	0.36 ± 0.03
		10	0.20 ± 0.06	0.16 ± 0.03	**0.40** ± 0.01	0.38 ± 0.03	0.22 ± 0.04	0.11 ± 0.04	**0.40** ± 0.01	0.39 ± 0.01
	1.0	2	0.06 ± 0.04	0.14 ± 0.06	**0.26** ± 0.03	0.21 ± 0.06	0.06 ± 0.04	0.08 ± 0.04	**0.26** ± 0.03	0.21 ± 0.06
		4	0.10 ± 0.04	0.17 ± 0.04	**0.34** ± 0.01	0.30 ± 0.02	0.10 ± 0.04	0.11 ± 0.06	**0.34** ± 0.01	0.30 ± 0.02
		6	0.13 ± 0.08	0.17 ± 0.04	**0.36** ± 0.02	0.32 ± 0.04	0.13 ± 0.08	0.12 ± 0.04	**0.36** ± 0.02	0.32 ± 0.04
		8	0.18 ± 0.08	0.18 ± 0.05	**0.39** ± 0.02	0.35 ± 0.03	0.18 ± 0.08	0.12 ± 0.04	**0.39** ± 0.02	0.35 ± 0.03
		10	0.21 ± 0.07	0.17 ± 0.04	**0.41** ± 0.01	0.39 ± 0.02	0.21 ± 0.07	0.10 ± 0.03	**0.41** ± 0.01	0.39 ± 0.02

Table 5: [Continued] Minimum global group representation $\text{minrep}_{\text{global}}(C, S')$ **in the clusterings returned by the baselines and our algorithms, for the** k**-Supplier objective (all values of** $n_f, k/t$**). Results are reported as** mean ± standard deviation **over ten independent runs.**

			Equitable requirement vector ($\vec{\beta}$)				Proportional requirement vector ($\vec{\beta}$)			
Dataset	n_f/n	k/t	FFT	FKSP	PILP	ALP	FFT	FKSP	PILP	ALP
MovieLens	0.2	2	0.00 ± 0.01	0.07 ± 0.05	**0.22** ± 0.05	0.17 ± 0.06	0.00 ± 0.00	0.06 ± 0.05	**0.22** ± 0.06	0.19 ± 0.07
		4	0.05 ± 0.07	0.10 ± 0.04	**0.32** ± 0.02	0.26 ± 0.06	0.08 ± 0.09	0.10 ± 0.05	**0.32** ± 0.03	0.25 ± 0.06
		6	0.06 ± 0.06	0.18 ± 0.04	**0.35** ± 0.01	0.31 ± 0.03	0.09 ± 0.06	0.07 ± 0.03	**0.35** ± 0.01	0.31 ± 0.03
		8	0.11 ± 0.05	0.15 ± 0.06	**0.36** ± 0.01	0.32 ± 0.04	0.10 ± 0.07	0.11 ± 0.04	**0.36** ± 0.01	0.32 ± 0.03
		10	0.13 ± 0.08	0.18 ± 0.03	**0.37** ± 0.01	0.34 ± 0.02	0.13 ± 0.06	0.12 ± 0.04	**0.37** ± 0.01	0.34 ± 0.02
	0.4	2	0.01 ± 0.03	0.06 ± 0.05	**0.21** ± 0.07	0.17 ± 0.08	0.01 ± 0.04	0.02 ± 0.03	**0.18** ± 0.08	0.14 ± 0.08
		4	0.03 ± 0.02	0.08 ± 0.04	**0.29** ± 0.04	0.23 ± 0.08	0.03 ± 0.03	0.09 ± 0.05	**0.30** ± 0.03	0.24 ± 0.05
		6	0.06 ± 0.06	0.10 ± 0.05	**0.34** ± 0.02	0.30 ± 0.05	0.10 ± 0.07	0.07 ± 0.03	**0.34** ± 0.02	0.28 ± 0.05
		8	0.09 ± 0.06	0.17 ± 0.02	**0.36** ± 0.01	0.31 ± 0.06	0.10 ± 0.07	0.08 ± 0.03	**0.36** ± 0.01	0.31 ± 0.04
		10	0.10 ± 0.06	0.14 ± 0.03	**0.37** ± 0.01	0.33 ± 0.03	0.14 ± 0.09	0.10 ± 0.04	**0.37** ± 0.01	0.33 ± 0.03
	0.6	2	0.01 ± 0.02	0.03 ± 0.02	**0.12** ± 0.07	0.08 ± 0.07	0.00 ± 0.01	0.03 ± 0.03	**0.14** ± 0.10	0.09 ± 0.11
		4	0.02 ± 0.02	0.09 ± 0.04	**0.26** ± 0.04	0.20 ± 0.05	0.03 ± 0.04	0.05 ± 0.05	**0.28** ± 0.05	0.20 ± 0.06
		6	0.06 ± 0.06	0.10 ± 0.05	**0.31** ± 0.04	0.25 ± 0.05	0.04 ± 0.02	0.08 ± 0.06	**0.32** ± 0.03	0.24 ± 0.05
		8	0.07 ± 0.04	0.13 ± 0.05	**0.35** ± 0.02	0.29 ± 0.06	0.07 ± 0.07	0.10 ± 0.05	**0.35** ± 0.01	0.29 ± 0.04
		10	0.08 ± 0.04	0.16 ± 0.05	**0.36** ± 0.02	0.33 ± 0.03	0.08 ± 0.04	0.09 ± 0.05	**0.35** ± 0.02	0.29 ± 0.06
	0.8	2	0.01 ± 0.01	0.03 ± 0.02	**0.10** ± 0.05	0.06 ± 0.04	0.01 ± 0.02	0.02 ± 0.02	**0.10** ± 0.05	0.07 ± 0.04
		4	0.04 ± 0.06	0.05 ± 0.02	**0.22** ± 0.07	0.16 ± 0.06	0.01 ± 0.02	0.05 ± 0.03	**0.20** ± 0.05	0.13 ± 0.06
		6	0.06 ± 0.05	0.07 ± 0.03	**0.28** ± 0.04	0.20 ± 0.03	0.04 ± 0.02	0.08 ± 0.05	**0.28** ± 0.03	0.18 ± 0.04
		8	0.06 ± 0.05	0.12 ± 0.04	**0.33** ± 0.02	0.25 ± 0.03	0.06 ± 0.05	0.07 ± 0.05	**0.31** ± 0.04	0.25 ± 0.06
		10	0.08 ± 0.05	0.09 ± 0.03	**0.34** ± 0.03	0.27 ± 0.07	0.07 ± 0.06	0.09 ± 0.05	**0.35** ± 0.02	0.29 ± 0.04
	1.0	2	0.01 ± 0.01	0.03 ± 0.02	**0.10** ± 0.05	0.06 ± 0.04	0.01 ± 0.01	0.01 ± 0.01	**0.10** ± 0.05	0.06 ± 0.04
		4	0.03 ± 0.05	0.07 ± 0.04	**0.20** ± 0.07	0.12 ± 0.07	0.03 ± 0.05	0.05 ± 0.02	**0.20** ± 0.07	0.12 ± 0.07
		6	0.04 ± 0.02	0.09 ± 0.03	**0.26** ± 0.06	0.18 ± 0.08	0.04 ± 0.02	0.09 ± 0.05	**0.26** ± 0.06	0.18 ± 0.08
		8	0.08 ± 0.07	0.11 ± 0.02	**0.29** ± 0.04	0.22 ± 0.06	0.08 ± 0.07	0.09 ± 0.05	**0.29** ± 0.04	0.22 ± 0.06
		10	0.10 ± 0.08	0.12 ± 0.05	**0.32** ± 0.04	0.26 ± 0.06	0.10 ± 0.08	0.06 ± 0.02	**0.32** ± 0.04	0.26 ± 0.06
Credit Card	0.2	2	0.00 ± 0.00	0.00 ± 0.00	**0.01** ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.01** ± 0.00	0.00 ± 0.00
		4	0.00 ± 0.00	0.00 ± 0.00	**0.06** ± 0.02	0.01 ± 0.01	0.00 ± 0.01	0.00 ± 0.00	**0.06** ± 0.03	0.01 ± 0.02
		6	0.00 ± 0.00	0.01 ± 0.01	**0.10** ± 0.02	0.04 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.10** ± 0.02	0.04 ± 0.03
		8	0.01 ± 0.02	0.02 ± 0.01	**0.12** ± 0.02	0.07 ± 0.04	0.01 ± 0.01	0.00 ± 0.01	**0.12** ± 0.02	0.06 ± 0.03
		10	0.01 ± 0.01	0.02 ± 0.01	**0.13** ± 0.02	0.07 ± 0.03	0.01 ± 0.01	0.01 ± 0.01	**0.13** ± 0.02	0.07 ± 0.04
	0.4	2	0.00 ± 0.00	0.00 ± 0.00	**0.01** ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.01** ± 0.00	0.00 ± 0.00
		4	0.00 ± 0.00	0.01 ± 0.01	**0.04** ± 0.02	0.01 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	**0.05** ± 0.03	0.01 ± 0.01
		6	0.00 ± 0.00	0.01 ± 0.00	**0.08** ± 0.03	0.03 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	**0.08** ± 0.02	0.03 ± 0.02
		8	0.00 ± 0.00	0.02 ± 0.01	**0.10** ± 0.01	0.04 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.11** ± 0.02	0.04 ± 0.03
		10	0.01 ± 0.01	0.02 ± 0.01	**0.12** ± 0.02	0.05 ± 0.03	0.01 ± 0.01	0.00 ± 0.01	**0.12** ± 0.01	0.06 ± 0.03
	0.6	2	0.00 ± 0.00	0.00 ± 0.00	**0.01** ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.01** ± 0.00	0.00 ± 0.00
		4	0.00 ± 0.00	0.00 ± 0.00	**0.04** ± 0.02	0.00 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	**0.04** ± 0.02	0.00 ± 0.01
		6	0.01 ± 0.01	0.01 ± 0.01	**0.07** ± 0.02	0.02 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.08** ± 0.03	0.03 ± 0.03
		8	0.01 ± 0.01	0.01 ± 0.01	**0.09** ± 0.02	0.04 ± 0.03	0.01 ± 0.01	0.00 ± 0.00	**0.10** ± 0.02	0.04 ± 0.03
		10	0.01 ± 0.00	0.01 ± 0.01	**0.11** ± 0.01	0.04 ± 0.02	0.01 ± 0.00	0.00 ± 0.01	**0.11** ± 0.02	0.03 ± 0.04
	0.8	2	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
		4	0.00 ± 0.00	0.00 ± 0.00	**0.03** ± 0.02	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.03** ± 0.02	0.00 ± 0.01
		6	0.00 ± 0.01	0.01 ± 0.00	**0.06** ± 0.02	0.02 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.07** ± 0.02	0.01 ± 0.01
		8	0.00 ± 0.01	0.01 ± 0.01	**0.09** ± 0.02	0.03 ± 0.02	0.01 ± 0.01	0.00 ± 0.00	**0.09** ± 0.03	0.03 ± 0.03
		10	0.00 ± 0.00	0.01 ± 0.01	**0.10** ± 0.01	0.03 ± 0.01	0.01 ± 0.01	0.00 ± 0.00	**0.10** ± 0.03	0.04 ± 0.04
	1.0	2	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
		4	0.00 ± 0.00	0.00 ± 0.00	**0.03** ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.03** ± 0.01	0.00 ± 0.00
		6	0.01 ± 0.01	0.00 ± 0.00	**0.05** ± 0.02	0.01 ± 0.01	0.01 ± 0.01	0.00 ± 0.00	**0.05** ± 0.02	0.01 ± 0.01
		8	0.01 ± 0.02	0.01 ± 0.01	**0.09** ± 0.03	0.03 ± 0.03	0.01 ± 0.02	0.01 ± 0.01	**0.09** ± 0.03	0.03 ± 0.03
		10	0.01 ± 0.01	0.01 ± 0.01	**0.10** ± 0.03	0.04 ± 0.04	0.01 ± 0.01	0.00 ± 0.00	**0.10** ± 0.03	0.04 ± 0.04

Table 5: [Continued] Minimum global group representation $\text{minrep}_{\text{global}}(C, S')$ **in the clusterings returned by the baselines and our algorithms, for the** k**-Supplier objective (all values of** $n_f, k/t$**). Results are reported as** mean ± standard deviation **over ten independent runs.**

			Equitable requirement vector ($\vec{\beta}$)				Proportional requirement vector ($\vec{\beta}$)			
Dataset	n_f/n	k/t	FFT	FKSP	PILP	ALP	FFT	FKSP	PILP	ALP
Bank	0.2	2	0.00 ± 0.00	0.04 ± 0.02	**0.16** ± 0.02	0.12 ± 0.03	0.01 ± 0.02	0.03 ± 0.02	**0.16** ± 0.02	0.09 ± 0.06
		4	0.01 ± 0.02	0.06 ± 0.02	**0.20** ± 0.01	0.15 ± 0.03	0.01 ± 0.02	0.02 ± 0.01	**0.20** ± 0.01	0.14 ± 0.04
		6	0.02 ± 0.03	0.08 ± 0.03	**0.21** ± 0.01	0.18 ± 0.02	0.02 ± 0.02	0.02 ± 0.02	**0.22** ± 0.01	0.18 ± 0.03
		8	0.01 ± 0.02	0.09 ± 0.02	**0.23** ± 0.00	0.20 ± 0.01	0.02 ± 0.02	0.02 ± 0.02	**0.22** ± 0.00	0.19 ± 0.02
		10	0.02 ± 0.02	0.09 ± 0.02	**0.23** ± 0.00	0.20 ± 0.02	0.02 ± 0.02	0.02 ± 0.02	**0.23** ± 0.00	0.20 ± 0.03
	0.4	2	0.00 ± 0.00	0.05 ± 0.04	**0.15** ± 0.02	0.08 ± 0.04	0.00 ± 0.00	0.03 ± 0.02	**0.15** ± 0.02	0.08 ± 0.05
		4	0.00 ± 0.01	0.07 ± 0.02	**0.20** ± 0.02	0.13 ± 0.05	0.01 ± 0.01	0.01 ± 0.01	**0.20** ± 0.01	0.14 ± 0.04
		6	0.01 ± 0.01	0.08 ± 0.02	**0.21** ± 0.01	0.16 ± 0.03	0.01 ± 0.01	0.02 ± 0.01	**0.21** ± 0.01	0.17 ± 0.03
		8	0.00 ± 0.00	0.09 ± 0.03	**0.22** ± 0.01	0.18 ± 0.03	0.00 ± 0.01	0.02 ± 0.02	**0.22** ± 0.01	0.18 ± 0.02
		10	0.02 ± 0.02	0.09 ± 0.02	**0.23** ± 0.00	0.19 ± 0.03	0.01 ± 0.01	0.02 ± 0.02	**0.23** ± 0.01	0.19 ± 0.03
	0.6	2	0.00 ± 0.00	0.06 ± 0.04	**0.15** ± 0.02	0.08 ± 0.04	0.00 ± 0.00	0.03 ± 0.02	**0.15** ± 0.02	0.08 ± 0.04
		4	0.00 ± 0.01	0.06 ± 0.02	**0.19** ± 0.02	0.13 ± 0.04	0.01 ± 0.01	0.02 ± 0.01	**0.20** ± 0.02	0.13 ± 0.04
		6	0.00 ± 0.00	0.08 ± 0.02	**0.21** ± 0.01	0.17 ± 0.03	0.01 ± 0.02	0.02 ± 0.01	**0.21** ± 0.01	0.17 ± 0.03
		8	0.01 ± 0.01	0.09 ± 0.03	**0.22** ± 0.01	0.19 ± 0.03	0.01 ± 0.01	0.02 ± 0.03	**0.22** ± 0.01	0.19 ± 0.03
		10	0.01 ± 0.01	0.08 ± 0.02	**0.22** ± 0.01	0.19 ± 0.02	0.02 ± 0.03	0.02 ± 0.02	**0.23** ± 0.01	0.20 ± 0.02
	0.8	2	0.00 ± 0.00	0.06 ± 0.02	**0.14** ± 0.02	0.08 ± 0.05	0.00 ± 0.01	0.04 ± 0.02	**0.14** ± 0.02	0.06 ± 0.05
		4	0.00 ± 0.01	0.06 ± 0.02	**0.19** ± 0.02	0.11 ± 0.05	0.00 ± 0.01	0.02 ± 0.01	**0.19** ± 0.02	0.13 ± 0.04
		6	0.00 ± 0.00	0.08 ± 0.02	**0.21** ± 0.01	0.17 ± 0.03	0.01 ± 0.01	0.02 ± 0.01	**0.21** ± 0.01	0.16 ± 0.03
		8	0.01 ± 0.01	0.07 ± 0.02	**0.22** ± 0.01	0.18 ± 0.02	0.00 ± 0.01	0.01 ± 0.01	**0.22** ± 0.01	0.17 ± 0.03
		10	0.02 ± 0.02	0.07 ± 0.02	**0.22** ± 0.01	0.18 ± 0.03	0.01 ± 0.01	0.02 ± 0.02	**0.22** ± 0.01	0.18 ± 0.04
	1.0	2	0.00 ± 0.00	0.04 ± 0.02	**0.14** ± 0.02	0.07 ± 0.05	0.00 ± 0.00	0.03 ± 0.02	**0.14** ± 0.02	0.07 ± 0.05
		4	0.00 ± 0.01	0.06 ± 0.02	**0.19** ± 0.02	0.12 ± 0.04	0.00 ± 0.01	0.01 ± 0.01	**0.19** ± 0.02	0.12 ± 0.04
		6	0.01 ± 0.01	0.09 ± 0.02	**0.21** ± 0.01	0.15 ± 0.04	0.01 ± 0.01	0.02 ± 0.01	**0.21** ± 0.01	0.15 ± 0.04
		8	0.01 ± 0.01	0.08 ± 0.03	**0.22** ± 0.01	0.18 ± 0.02	0.01 ± 0.01	0.01 ± 0.01	**0.22** ± 0.01	0.18 ± 0.02
		10	0.02 ± 0.03	0.06 ± 0.02	**0.22** ± 0.01	0.18 ± 0.03	0.02 ± 0.03	0.03 ± 0.02	**0.22** ± 0.01	0.18 ± 0.03
Adult	0.2	2	0.00 ± 0.00	0.01 ± 0.01	**0.10** ± 0.01	0.04 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	**0.10** ± 0.01	0.04 ± 0.02
		4	0.00 ± 0.00	0.01 ± 0.00	**0.14** ± 0.01	0.10 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.14** ± 0.01	0.10 ± 0.02
		6	0.00 ± 0.00	0.01 ± 0.00	**0.16** ± 0.00	0.13 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	**0.16** ± 0.01	0.13 ± 0.02
		8	0.00 ± 0.00	0.02 ± 0.01	**0.17** ± 0.00	0.14 ± 0.01	0.01 ± 0.01	0.00 ± 0.00	**0.17** ± 0.00	0.14 ± 0.01
		10	0.01 ± 0.01	0.02 ± 0.01	**0.18** ± 0.00	0.15 ± 0.01	0.01 ± 0.01	0.00 ± 0.00	**0.18** ± 0.00	0.16 ± 0.01
	0.4	2	0.00 ± 0.00	0.01 ± 0.01	**0.10** ± 0.01	0.04 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.10** ± 0.01	0.04 ± 0.03
		4	0.00 ± 0.00	0.01 ± 0.00	**0.14** ± 0.01	0.08 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	**0.14** ± 0.01	0.09 ± 0.02
		6	0.00 ± 0.00	0.02 ± 0.01	**0.16** ± 0.00	0.12 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.16** ± 0.00	0.12 ± 0.02
		8	0.00 ± 0.00	0.01 ± 0.00	**0.17** ± 0.00	0.14 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.17** ± 0.00	0.14 ± 0.01
		10	0.01 ± 0.00	0.02 ± 0.00	**0.18** ± 0.00	0.15 ± 0.01	0.00 ± 0.01	0.00 ± 0.00	**0.18** ± 0.00	0.15 ± 0.01
	0.6	2	0.00 ± 0.00	0.01 ± 0.00	**0.10** ± 0.01	0.03 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.09** ± 0.01	0.03 ± 0.02
		4	0.00 ± 0.00	0.01 ± 0.00	**0.14** ± 0.01	0.08 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	**0.14** ± 0.01	0.08 ± 0.04
		6	0.00 ± 0.00	0.01 ± 0.01	**0.16** ± 0.00	0.10 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	**0.16** ± 0.01	0.11 ± 0.03
		8	0.00 ± 0.01	0.01 ± 0.01	**0.17** ± 0.00	0.12 ± 0.02	0.00 ± 0.01	0.00 ± 0.00	**0.17** ± 0.00	0.14 ± 0.01
		10	0.01 ± 0.00	0.01 ± 0.01	**0.18** ± 0.00	0.15 ± 0.01	0.01 ± 0.01	0.00 ± 0.00	**0.18** ± 0.00	0.16 ± 0.01
	0.8	2	0.00 ± 0.00	0.01 ± 0.00	**0.09** ± 0.01	0.02 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.09** ± 0.01	0.02 ± 0.02
		4	0.00 ± 0.00	0.01 ± 0.00	**0.13** ± 0.01	0.06 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	**0.13** ± 0.01	0.07 ± 0.03
		6	0.00 ± 0.00	0.01 ± 0.01	**0.15** ± 0.01	0.10 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.15** ± 0.01	0.10 ± 0.02
		8	0.00 ± 0.01	0.01 ± 0.01	**0.17** ± 0.00	0.12 ± 0.03	0.01 ± 0.01	0.00 ± 0.00	**0.17** ± 0.00	0.12 ± 0.02
		10	0.01 ± 0.00	0.02 ± 0.01	**0.18** ± 0.00	0.14 ± 0.02	0.01 ± 0.01	0.00 ± 0.00	**0.18** ± 0.00	0.13 ± 0.02
	1.0	2	0.00 ± 0.00	0.01 ± 0.00	**0.09** ± 0.01	0.03 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.09** ± 0.01	0.03 ± 0.02
		4	0.00 ± 0.00	0.01 ± 0.00	**0.13** ± 0.01	0.07 ± 0.04	0.00 ± 0.00	0.00 ± 0.00	**0.13** ± 0.01	0.07 ± 0.04
		6	0.00 ± 0.00	0.02 ± 0.01	**0.15** ± 0.01	0.10 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	**0.15** ± 0.01	0.10 ± 0.03
		8	0.00 ± 0.00	0.01 ± 0.01	**0.17** ± 0.01	0.12 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	**0.17** ± 0.01	0.12 ± 0.03
		10	0.01 ± 0.01	0.02 ± 0.01	**0.18** ± 0.01	0.14 ± 0.02	0.01 ± 0.01	0.00 ± 0.00	**0.18** ± 0.01	0.14 ± 0.02

Table 6: Minimum local group representation $\text{minrep}_{\text{local}}(C, S')$ **in the clusterings returned by the baselines and our algorithms, for the** k**-Supplier objective (all values of** $n_f, k/t$**). Results are reported as** mean ± standard deviation **over ten independent runs.**

			Equitable requirement vector ($\vec{\beta}$)				Proportional requirement vector ($\vec{\beta}$)			
Dataset	n_f/n	k/t	FFT	FKSP	PILP	ALP	FFT	FKSP	PILP	ALP
Heart	0.2	2	0.06 ± 0.10	0.26 ± 0.15	**0.48** ± 0.10	0.46 ± 0.09	0.10 ± 0.15	0.14 ± 0.08	**0.46** ± 0.14	0.45 ± 0.14
		4	0.34 ± 0.27	0.57 ± 0.16	**0.97** ± 0.18	0.95 ± 0.17	0.29 ± 0.22	0.40 ± 0.18	**0.81** ± 0.18	0.80 ± 0.18
		6	0.76 ± 0.45	0.67 ± 0.12	**1.40** ± 0.36	**1.40** ± 0.36	0.56 ± 0.41	0.33 ± 0.22	**1.10** ± 0.31	**1.10** ± 0.31
		8	0.91 ± 0.52	0.77 ± 0.16	**1.62** ± 0.33	**1.62** ± 0.33	0.66 ± 0.43	0.55 ± 0.27	**1.43** ± 0.32	**1.43** ± 0.32
		10	1.06 ± 0.46	0.94 ± 0.28	**1.92** ± 0.37	**1.92** ± 0.37	1.04 ± 0.46	0.59 ± 0.20	**1.66** ± 0.39	**1.66** ± 0.39
	0.4	2	0.18 ± 0.19	0.31 ± 0.12	**0.51** ± 0.03	0.50 ± 0.03	0.11 ± 0.14	0.14 ± 0.10	**0.52** ± 0.04	**0.52** ± 0.04
		4	0.20 ± 0.19	0.42 ± 0.24	**1.08** ± 0.24	1.07 ± 0.24	0.24 ± 0.24	0.32 ± 0.21	**1.16** ± 0.25	1.13 ± 0.24
		6	0.69 ± 0.60	0.90 ± 0.28	**1.88** ± 0.55	1.87 ± 0.54	0.63 ± 0.54	0.47 ± 0.22	**1.66** ± 0.47	**1.66** ± 0.46
		8	1.12 ± 0.40	0.87 ± 0.30	**2.21** ± 0.36	2.20 ± 0.36	0.94 ± 0.68	0.63 ± 0.17	**2.24** ± 0.40	2.23 ± 0.39
		10	1.17 ± 0.49	1.44 ± 0.44	**2.42** ± 0.36	2.41 ± 0.36	1.00 ± 0.59	0.87 ± 0.40	**2.41** ± 0.54	2.40 ± 0.54
	0.6	2	0.15 ± 0.13	0.31 ± 0.07	**0.54** ± 0.05	0.51 ± 0.05	0.14 ± 0.19	0.13 ± 0.09	**0.54** ± 0.07	0.52 ± 0.06
		4	0.26 ± 0.16	0.57 ± 0.19	**1.29** ± 0.16	1.28 ± 0.17	0.22 ± 0.24	0.39 ± 0.16	**1.30** ± 0.14	1.27 ± 0.14
		6	0.66 ± 0.31	0.89 ± 0.27	**2.10** ± 0.30	2.09 ± 0.29	0.37 ± 0.26	0.55 ± 0.20	**1.86** ± 0.23	**1.86** ± 0.23
		8	0.89 ± 0.53	1.15 ± 0.54	**2.54** ± 0.31	2.52 ± 0.30	0.83 ± 0.39	0.74 ± 0.24	**2.78** ± 0.45	**2.78** ± 0.44
		10	1.13 ± 0.65	1.33 ± 0.68	**2.87** ± 0.51	**2.87** ± 0.49	0.80 ± 0.70	0.94 ± 0.48	**2.80** ± 0.45	**2.80** ± 0.45
	0.8	2	0.12 ± 0.11	0.30 ± 0.12	**0.54** ± 0.04	0.52 ± 0.05	0.07 ± 0.09	0.13 ± 0.09	**0.54** ± 0.04	0.53 ± 0.05
		4	0.26 ± 0.28	0.53 ± 0.20	**1.34** ± 0.10	1.30 ± 0.10	0.28 ± 0.27	0.42 ± 0.16	**1.30** ± 0.10	1.25 ± 0.12
		6	0.64 ± 0.41	0.94 ± 0.28	**2.10** ± 0.25	2.08 ± 0.24	0.49 ± 0.42	0.48 ± 0.16	**2.12** ± 0.22	2.11 ± 0.21
		8	0.71 ± 0.52	1.12 ± 0.29	**2.93** ± 0.34	2.92 ± 0.33	0.88 ± 0.50	0.76 ± 0.28	**2.71** ± 0.42	2.70 ± 0.42
		10	1.09 ± 0.57	1.69 ± 0.42	**3.21** ± 0.40	**3.21** ± 0.40	1.12 ± 0.64	0.98 ± 0.48	**3.47** ± 0.44	**3.47** ± 0.44
	1.0	2	0.12 ± 0.16	0.31 ± 0.12	**0.54** ± 0.04	0.53 ± 0.05	0.12 ± 0.16	0.11 ± 0.08	**0.54** ± 0.04	0.53 ± 0.05
		4	0.22 ± 0.24	0.54 ± 0.24	**1.31** ± 0.10	1.28 ± 0.10	0.22 ± 0.24	0.35 ± 0.20	**1.31** ± 0.10	1.28 ± 0.10
		6	0.59 ± 0.38	0.87 ± 0.29	**2.20** ± 0.23	2.17 ± 0.22	0.59 ± 0.38	0.66 ± 0.26	**2.20** ± 0.23	2.17 ± 0.22
		8	0.82 ± 0.53	1.24 ± 0.43	**3.04** ± 0.47	3.03 ± 0.46	0.82 ± 0.53	0.74 ± 0.20	**3.04** ± 0.47	3.03 ± 0.46
		10	1.09 ± 0.61	1.67 ± 0.38	**3.52** ± 0.41	**3.52** ± 0.41	1.09 ± 0.61	1.19 ± 0.37	**3.52** ± 0.41	**3.52** ± 0.41
Student	0.2	2	0.09 ± 0.12	0.20 ± 0.06	**0.46** ± 0.04	0.44 ± 0.04	0.07 ± 0.08	0.10 ± 0.02	**0.47** ± 0.04	0.46 ± 0.05
		4	0.50 ± 0.34	0.48 ± 0.24	**1.12** ± 0.11	1.10 ± 0.10	0.49 ± 0.16	0.21 ± 0.10	**1.13** ± 0.14	1.12 ± 0.15
		6	0.62 ± 0.35	0.75 ± 0.29	**1.71** ± 0.28	1.69 ± 0.27	0.72 ± 0.34	0.36 ± 0.20	**1.63** ± 0.31	1.62 ± 0.30
		8	1.18 ± 0.41	0.98 ± 0.33	**2.54** ± 0.30	2.50 ± 0.30	0.99 ± 0.26	0.64 ± 0.27	**2.17** ± 0.31	2.16 ± 0.30
		10	1.61 ± 0.48	1.36 ± 0.32	**2.81** ± 0.56	2.80 ± 0.55	1.33 ± 0.50	0.59 ± 0.30	**2.62** ± 0.29	2.61 ± 0.29
	0.4	2	0.05 ± 0.11	0.20 ± 0.11	**0.48** ± 0.04	0.47 ± 0.04	0.10 ± 0.14	0.14 ± 0.07	**0.47** ± 0.04	0.46 ± 0.04
		4	0.39 ± 0.23	0.55 ± 0.15	**1.27** ± 0.09	1.25 ± 0.10	0.42 ± 0.35	0.22 ± 0.13	**1.24** ± 0.18	1.22 ± 0.19
		6	0.67 ± 0.23	0.73 ± 0.23	**1.88** ± 0.14	1.84 ± 0.15	0.74 ± 0.16	0.33 ± 0.21	**1.96** ± 0.15	1.92 ± 0.17
		8	1.40 ± 0.41	1.15 ± 0.21	**2.88** ± 0.23	2.85 ± 0.23	1.17 ± 0.60	0.59 ± 0.21	**2.78** ± 0.20	2.74 ± 0.21
		10	1.58 ± 0.56	1.35 ± 0.45	**3.69** ± 0.26	3.66 ± 0.26	1.62 ± 0.68	0.65 ± 0.30	**3.54** ± 0.49	3.52 ± 0.49
	0.6	2	0.08 ± 0.14	0.22 ± 0.06	**0.47** ± 0.06	0.46 ± 0.06	0.18 ± 0.15	0.09 ± 0.03	**0.48** ± 0.05	**0.48** ± 0.06
		4	0.28 ± 0.25	0.53 ± 0.22	**1.19** ± 0.12	1.17 ± 0.12	0.43 ± 0.35	0.20 ± 0.12	**1.22** ± 0.14	1.19 ± 0.15
		6	0.77 ± 0.39	0.79 ± 0.28	**2.15** ± 0.12	2.12 ± 0.13	0.91 ± 0.32	0.44 ± 0.19	**2.07** ± 0.20	2.04 ± 0.21
		8	1.34 ± 0.53	1.06 ± 0.20	**2.97** ± 0.23	2.92 ± 0.21	1.25 ± 0.55	0.48 ± 0.16	**2.87** ± 0.25	2.84 ± 0.27
		10	2.02 ± 0.65	1.43 ± 0.42	**4.01** ± 0.33	3.97 ± 0.33	1.82 ± 0.62	0.65 ± 0.27	**3.87** ± 0.27	3.83 ± 0.26
	0.8	2	0.09 ± 0.09	0.21 ± 0.04	**0.47** ± 0.04	0.44 ± 0.05	0.15 ± 0.12	0.09 ± 0.03	**0.48** ± 0.04	0.46 ± 0.04
		4	0.40 ± 0.18	0.53 ± 0.20	**1.21** ± 0.07	1.20 ± 0.07	0.31 ± 0.20	0.27 ± 0.14	**1.21** ± 0.10	1.16 ± 0.12
		6	0.80 ± 0.31	0.81 ± 0.20	**2.02** ± 0.20	2.00 ± 0.18	0.84 ± 0.33	0.46 ± 0.23	**2.01** ± 0.12	1.98 ± 0.13
		8	1.12 ± 0.34	1.25 ± 0.28	**2.92** ± 0.23	2.89 ± 0.22	1.25 ± 0.39	0.60 ± 0.24	**2.97** ± 0.13	2.93 ± 0.15
		10	1.89 ± 0.39	1.56 ± 0.48	**4.03** ± 0.24	3.97 ± 0.24	2.03 ± 0.80	0.78 ± 0.39	**4.06** ± 0.29	4.01 ± 0.31
	1.0	2	0.17 ± 0.11	0.26 ± 0.08	**0.48** ± 0.04	0.47 ± 0.04	0.17 ± 0.11	0.10 ± 0.04	**0.48** ± 0.04	0.47 ± 0.04
		4	0.38 ± 0.13	0.54 ± 0.19	**1.19** ± 0.08	1.16 ± 0.10	0.38 ± 0.13	0.25 ± 0.14	**1.19** ± 0.08	1.16 ± 0.10
		6	0.64 ± 0.27	0.81 ± 0.21	**1.89** ± 0.12	1.85 ± 0.12	0.64 ± 0.27	0.48 ± 0.19	**1.89** ± 0.12	1.85 ± 0.12
		8	1.13 ± 0.50	1.13 ± 0.25	**2.88** ± 0.20	2.84 ± 0.20	1.13 ± 0.50	0.58 ± 0.19	**2.88** ± 0.20	2.84 ± 0.20
		10	1.88 ± 0.58	1.58 ± 0.52	**3.97** ± 0.13	3.91 ± 0.16	1.88 ± 0.58	0.70 ± 0.24	**3.97** ± 0.13	3.91 ± 0.16

Table 6: [Continued] Minimum local group representation $\text{minrep}_{\text{local}}(C, S')$ in the clusterings returned by the baselines and our algorithms, for the k-Supplier objective (all values of $n_f, k/t$). Results are reported as mean ± standard deviation **over ten independent runs.**

			Equitable requirement vector ($\vec{\beta}$)				Proportional requirement vector ($\vec{\beta}$)			
Dataset	n_f/n	k/t	FFT	FKSP	PILP	ALP	FFT	FKSP	PILP	ALP
MovieLens	0.2	2	0.01 ± 0.04	0.23 ± 0.07	**0.51** ± 0.04	0.48 ± 0.06	0.04 ± 0.12	0.10 ± 0.03	**0.51** ± 0.04	0.49 ± 0.05
		4	0.23 ± 0.22	0.58 ± 0.14	**1.20** ± 0.11	1.18 ± 0.12	0.26 ± 0.20	0.25 ± 0.08	**1.13** ± 0.10	1.10 ± 0.11
		6	0.61 ± 0.40	0.79 ± 0.18	**1.84** ± 0.17	1.81 ± 0.17	0.45 ± 0.32	0.38 ± 0.13	**1.82** ± 0.14	1.77 ± 0.16
		8	0.89 ± 0.30	1.08 ± 0.23	**2.40** ± 0.17	2.37 ± 0.15	0.75 ± 0.40	0.54 ± 0.16	**2.32** ± 0.11	2.29 ± 0.11
		10	1.01 ± 0.31	1.29 ± 0.26	**2.98** ± 0.20	2.94 ± 0.19	0.93 ± 0.25	0.62 ± 0.14	**3.03** ± 0.18	2.98 ± 0.19
	0.4	2	0.03 ± 0.07	0.23 ± 0.12	**0.53** ± 0.04	0.52 ± 0.04	0.07 ± 0.12	0.11 ± 0.06	**0.51** ± 0.04	0.49 ± 0.03
		4	0.25 ± 0.18	0.55 ± 0.12	**1.18** ± 0.09	1.16 ± 0.09	0.33 ± 0.25	0.31 ± 0.10	**1.21** ± 0.08	1.15 ± 0.09
		6	0.42 ± 0.25	0.74 ± 0.33	**1.88** ± 0.15	1.85 ± 0.15	0.63 ± 0.22	0.45 ± 0.11	**1.84** ± 0.15	1.79 ± 0.12
		8	0.84 ± 0.30	1.18 ± 0.35	**2.53** ± 0.24	2.49 ± 0.25	1.01 ± 0.55	0.61 ± 0.17	**2.61** ± 0.35	2.56 ± 0.35
		10	1.02 ± 0.40	1.18 ± 0.30	**3.23** ± 0.20	3.19 ± 0.18	1.17 ± 0.53	0.73 ± 0.30	**3.19** ± 0.26	3.15 ± 0.26
	0.6	2	0.09 ± 0.15	0.26 ± 0.15	**0.52** ± 0.06	0.51 ± 0.06	0.04 ± 0.12	0.08 ± 0.04	**0.53** ± 0.06	0.52 ± 0.06
		4	0.25 ± 0.19	0.54 ± 0.11	**1.22** ± 0.10	1.19 ± 0.10	0.24 ± 0.19	0.30 ± 0.15	**1.19** ± 0.09	1.16 ± 0.09
		6	0.51 ± 0.31	0.87 ± 0.21	**1.90** ± 0.14	1.86 ± 0.15	0.43 ± 0.22	0.48 ± 0.19	**1.96** ± 0.13	1.90 ± 0.14
		8	0.76 ± 0.22	1.15 ± 0.37	**2.69** ± 0.25	2.66 ± 0.27	0.65 ± 0.42	0.60 ± 0.14	**2.54** ± 0.22	2.48 ± 0.21
		10	1.12 ± 0.46	1.29 ± 0.29	**3.37** ± 0.24	3.32 ± 0.23	1.09 ± 0.40	0.64 ± 0.21	**3.39** ± 0.20	3.35 ± 0.20
	0.8	2	0.13 ± 0.12	0.25 ± 0.10	**0.54** ± 0.03	0.52 ± 0.03	0.18 ± 0.13	0.11 ± 0.05	**0.53** ± 0.05	0.52 ± 0.05
		4	0.34 ± 0.22	0.59 ± 0.10	**1.25** ± 0.10	1.21 ± 0.09	0.28 ± 0.20	0.27 ± 0.06	**1.25** ± 0.11	1.21 ± 0.11
		6	0.61 ± 0.31	0.89 ± 0.26	**1.89** ± 0.07	1.85 ± 0.09	0.69 ± 0.20	0.56 ± 0.23	**2.02** ± 0.10	1.98 ± 0.11
		8	0.89 ± 0.33	1.22 ± 0.23	**2.67** ± 0.15	2.61 ± 0.16	0.78 ± 0.33	0.64 ± 0.21	**2.65** ± 0.17	2.58 ± 0.18
		10	1.12 ± 0.46	1.49 ± 0.46	**3.51** ± 0.35	3.44 ± 0.36	1.02 ± 0.39	0.67 ± 0.21	**3.33** ± 0.36	3.29 ± 0.35
	1.0	2	0.15 ± 0.16	0.38 ± 0.10	**0.54** ± 0.05	**0.54** ± 0.05	0.15 ± 0.16	0.15 ± 0.11	**0.54** ± 0.05	**0.54** ± 0.05
		4	0.29 ± 0.21	0.61 ± 0.14	**1.27** ± 0.10	1.24 ± 0.11	0.29 ± 0.21	0.35 ± 0.15	**1.27** ± 0.10	1.24 ± 0.11
		6	0.66 ± 0.31	0.99 ± 0.23	**1.96** ± 0.15	1.92 ± 0.16	0.66 ± 0.31	0.58 ± 0.19	**1.96** ± 0.15	1.92 ± 0.16
		8	0.88 ± 0.26	1.41 ± 0.29	**2.66** ± 0.20	2.63 ± 0.19	0.88 ± 0.26	0.85 ± 0.24	**2.66** ± 0.20	2.63 ± 0.19
		10	1.14 ± 0.22	1.64 ± 0.39	**3.44** ± 0.22	3.40 ± 0.22	1.14 ± 0.22	0.95 ± 0.38	**3.44** ± 0.22	3.40 ± 0.22
Credit Card	0.2	2	0.00 ± 0.01	0.06 ± 0.04	**0.22** ± 0.05	0.19 ± 0.06	0.00 ± 0.00	0.01 ± 0.02	**0.19** ± 0.04	0.18 ± 0.05
		4	0.05 ± 0.04	0.08 ± 0.05	**0.49** ± 0.10	0.47 ± 0.11	0.03 ± 0.05	0.03 ± 0.03	**0.51** ± 0.13	0.49 ± 0.14
		6	0.10 ± 0.15	0.13 ± 0.08	**0.79** ± 0.17	0.78 ± 0.17	0.13 ± 0.18	0.03 ± 0.06	**0.87** ± 0.32	0.85 ± 0.31
		8	0.19 ± 0.18	0.28 ± 0.21	**1.10** ± 0.27	1.07 ± 0.26	0.28 ± 0.19	0.06 ± 0.09	**1.25** ± 0.32	1.23 ± 0.31
		10	0.34 ± 0.27	0.29 ± 0.21	**1.46** ± 0.39	1.44 ± 0.38	0.31 ± 0.29	0.04 ± 0.09	**1.48** ± 0.34	1.46 ± 0.34
	0.4	2	0.01 ± 0.02	0.06 ± 0.03	**0.22** ± 0.03	0.21 ± 0.03	0.00 ± 0.00	0.01 ± 0.02	**0.21** ± 0.03	**0.21** ± 0.03
		4	0.07 ± 0.09	0.07 ± 0.02	**0.78** ± 0.18	0.76 ± 0.16	0.09 ± 0.11	0.02 ± 0.02	**0.74** ± 0.20	0.72 ± 0.20
		6	0.08 ± 0.07	0.16 ± 0.09	**0.96** ± 0.16	0.94 ± 0.15	0.31 ± 0.26	0.02 ± 0.04	**1.35** ± 0.29	1.31 ± 0.28
		8	0.18 ± 0.23	0.23 ± 0.17	**1.40** ± 0.30	1.38 ± 0.29	0.29 ± 0.29	0.06 ± 0.05	**1.59** ± 0.38	1.56 ± 0.36
		10	0.34 ± 0.21	0.28 ± 0.19	**1.70** ± 0.34	1.68 ± 0.33	0.34 ± 0.29	0.03 ± 0.03	**1.79** ± 0.33	1.78 ± 0.32
	0.6	2	0.01 ± 0.02	0.05 ± 0.03	**0.25** ± 0.06	0.22 ± 0.10	0.02 ± 0.04	0.01 ± 0.02	**0.25** ± 0.08	0.23 ± 0.08
		4	0.12 ± 0.11	0.10 ± 0.05	**0.89** ± 0.18	0.86 ± 0.16	0.12 ± 0.16	0.00 ± 0.01	**0.86** ± 0.28	0.85 ± 0.28
		6	0.17 ± 0.15	0.18 ± 0.14	**1.18** ± 0.26	1.16 ± 0.24	0.53 ± 0.39	0.05 ± 0.09	**1.53** ± 0.34	1.50 ± 0.33
		8	0.28 ± 0.36	0.28 ± 0.21	**1.66** ± 0.45	1.65 ± 0.45	0.45 ± 0.45	0.06 ± 0.13	**1.90** ± 0.30	1.86 ± 0.29
		10	0.50 ± 0.37	0.25 ± 0.16	**2.29** ± 0.51	2.26 ± 0.48	0.51 ± 0.43	0.01 ± 0.02	**2.16** ± 0.32	2.13 ± 0.31
	0.8	2	0.01 ± 0.03	0.05 ± 0.03	**0.27** ± 0.07	0.26 ± 0.07	0.02 ± 0.03	0.02 ± 0.02	**0.23** ± 0.04	**0.23** ± 0.04
		4	0.15 ± 0.14	0.10 ± 0.08	**1.08** ± 0.16	1.04 ± 0.16	0.14 ± 0.16	0.01 ± 0.02	**1.00** ± 0.15	0.98 ± 0.13
		6	0.29 ± 0.24	0.27 ± 0.21	**1.49** ± 0.36	1.46 ± 0.36	0.56 ± 0.32	0.19 ± 0.32	**1.68** ± 0.10	1.64 ± 0.10
		8	0.42 ± 0.40	0.31 ± 0.20	**1.89** ± 0.43	1.83 ± 0.40	0.55 ± 0.38	0.03 ± 0.04	**2.16** ± 0.33	2.12 ± 0.30
		10	0.90 ± 0.35	0.32 ± 0.17	**2.74** ± 0.38	2.68 ± 0.39	0.77 ± 0.45	0.09 ± 0.13	**2.63** ± 0.40	2.60 ± 0.39
	1.0	2	0.02 ± 0.03	0.05 ± 0.04	**0.25** ± 0.05	**0.25** ± 0.05	0.02 ± 0.03	0.01 ± 0.02	**0.25** ± 0.05	**0.25** ± 0.05
		4	0.25 ± 0.19	0.23 ± 0.13	**1.05** ± 0.13	1.03 ± 0.12	0.25 ± 0.19	0.05 ± 0.07	**1.05** ± 0.13	1.03 ± 0.12
		6	0.49 ± 0.22	0.42 ± 0.17	**1.65** ± 0.14	1.59 ± 0.13	0.49 ± 0.22	0.27 ± 0.17	**1.65** ± 0.14	1.59 ± 0.13
		8	0.69 ± 0.23	0.68 ± 0.22	**2.33** ± 0.12	2.30 ± 0.13	0.69 ± 0.23	0.51 ± 0.22	**2.33** ± 0.12	2.30 ± 0.13
		10	1.04 ± 0.14	1.11 ± 0.31	**2.94** ± 0.17	2.90 ± 0.17	1.04 ± 0.14	0.87 ± 0.26	**2.94** ± 0.17	2.90 ± 0.17

Table 6: [Continued] Minimum local group representation $\text{minrep}_{\text{local}}(C, S')$ in the clusterings returned by the baselines and our algorithms, for the k-Supplier objective (all values of $n_f, k/t$). Results are reported as mean ± standard deviation **over ten independent runs.**

			Equitable requirement vector ($\vec{\beta}$)				Proportional requirement vector ($\vec{\beta}$)			
Dataset	n_f/n	k/t	FFT	FKSP	PILP	ALP	FFT	FKSP	PILP	ALP
Bank	0.2	2	0.00 ± 0.00	0.06 ± 0.02	**0.18** ± 0.01	0.16 ± 0.03	0.01 ± 0.02	0.03 ± 0.01	**0.18** ± 0.01	0.17 ± 0.02
		4	0.02 ± 0.02	0.13 ± 0.03	**0.40** ± 0.02	0.38 ± 0.02	0.02 ± 0.03	0.03 ± 0.02	**0.41** ± 0.02	0.40 ± 0.02
		6	0.03 ± 0.03	0.19 ± 0.04	**0.66** ± 0.03	0.63 ± 0.04	0.03 ± 0.04	0.05 ± 0.02	**0.67** ± 0.02	0.65 ± 0.02
		8	0.04 ± 0.04	0.24 ± 0.03	**0.90** ± 0.02	0.86 ± 0.07	0.04 ± 0.04	0.05 ± 0.03	**0.89** ± 0.03	0.86 ± 0.02
		10	0.05 ± 0.04	0.31 ± 0.07	**1.14** ± 0.04	1.12 ± 0.04	0.05 ± 0.06	0.06 ± 0.03	**1.12** ± 0.05	1.10 ± 0.05
	0.4	2	0.00 ± 0.00	0.06 ± 0.02	**0.17** ± 0.01	0.16 ± 0.02	0.00 ± 0.00	0.03 ± 0.01	**0.18** ± 0.01	0.16 ± 0.03
		4	0.01 ± 0.02	0.12 ± 0.02	**0.41** ± 0.01	0.39 ± 0.01	0.01 ± 0.02	0.03 ± 0.02	**0.41** ± 0.02	0.39 ± 0.04
		6	0.02 ± 0.03	0.18 ± 0.04	**0.66** ± 0.03	0.65 ± 0.03	0.02 ± 0.03	0.05 ± 0.03	**0.67** ± 0.03	0.65 ± 0.04
		8	0.02 ± 0.03	0.24 ± 0.04	**0.91** ± 0.04	0.88 ± 0.06	0.03 ± 0.05	0.05 ± 0.03	**0.92** ± 0.03	0.89 ± 0.03
		10	0.06 ± 0.05	0.28 ± 0.06	**1.18** ± 0.04	1.16 ± 0.04	0.06 ± 0.06	0.06 ± 0.04	**1.15** ± 0.04	1.13 ± 0.04
	0.6	2	0.00 ± 0.00	0.06 ± 0.02	**0.17** ± 0.01	0.15 ± 0.02	0.00 ± 0.00	0.03 ± 0.01	**0.17** ± 0.01	0.16 ± 0.02
		4	0.02 ± 0.03	0.12 ± 0.02	**0.41** ± 0.01	0.40 ± 0.01	0.01 ± 0.02	0.03 ± 0.01	**0.41** ± 0.01	0.39 ± 0.03
		6	0.02 ± 0.02	0.18 ± 0.04	**0.66** ± 0.02	0.63 ± 0.04	0.02 ± 0.02	0.05 ± 0.02	**0.66** ± 0.02	0.63 ± 0.03
		8	0.02 ± 0.04	0.23 ± 0.04	**0.93** ± 0.03	0.91 ± 0.04	0.03 ± 0.04	0.04 ± 0.03	**0.92** ± 0.03	0.90 ± 0.03
		10	0.04 ± 0.03	0.27 ± 0.06	**1.17** ± 0.05	1.13 ± 0.07	0.05 ± 0.03	0.07 ± 0.03	**1.17** ± 0.04	1.15 ± 0.05
	0.8	2	0.00 ± 0.00	0.07 ± 0.02	**0.17** ± 0.01	0.16 ± 0.02	0.00 ± 0.01	0.03 ± 0.01	**0.17** ± 0.01	0.16 ± 0.02
		4	0.01 ± 0.02	0.12 ± 0.02	**0.42** ± 0.01	0.41 ± 0.01	0.01 ± 0.02	0.03 ± 0.02	**0.41** ± 0.01	0.39 ± 0.03
		6	0.02 ± 0.02	0.17 ± 0.04	**0.66** ± 0.02	0.63 ± 0.04	0.03 ± 0.04	0.06 ± 0.03	**0.66** ± 0.02	0.63 ± 0.04
		8	0.03 ± 0.05	0.24 ± 0.03	**0.93** ± 0.03	0.91 ± 0.04	0.03 ± 0.04	0.05 ± 0.03	**0.92** ± 0.03	0.91 ± 0.04
		10	0.07 ± 0.05	0.28 ± 0.04	**1.18** ± 0.04	1.16 ± 0.05	0.04 ± 0.04	0.06 ± 0.02	**1.17** ± 0.04	1.15 ± 0.04
	1.0	2	0.00 ± 0.00	0.07 ± 0.01	**0.17** ± 0.01	0.15 ± 0.03	0.00 ± 0.00	0.03 ± 0.01	**0.17** ± 0.01	0.15 ± 0.03
		4	0.02 ± 0.02	0.11 ± 0.02	**0.41** ± 0.01	0.40 ± 0.02	0.02 ± 0.02	0.03 ± 0.01	**0.41** ± 0.01	0.40 ± 0.02
		6	0.03 ± 0.03	0.17 ± 0.02	**0.66** ± 0.02	0.63 ± 0.05	0.03 ± 0.03	0.06 ± 0.02	**0.66** ± 0.02	0.63 ± 0.05
		8	0.04 ± 0.05	0.24 ± 0.04	**0.92** ± 0.03	0.89 ± 0.04	0.04 ± 0.05	0.05 ± 0.03	**0.92** ± 0.03	0.89 ± 0.04
		10	0.06 ± 0.06	0.28 ± 0.03	**1.16** ± 0.03	1.14 ± 0.04	0.06 ± 0.06	0.08 ± 0.03	**1.16** ± 0.03	1.14 ± 0.04
Adult	0.2	2	0.00 ± 0.00	0.01 ± 0.00	**0.07** ± 0.01	0.00 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.07** ± 0.01	0.01 ± 0.02
		4	0.00 ± 0.00	0.02 ± 0.01	**0.19** ± 0.01	0.13 ± 0.06	0.00 ± 0.00	0.00 ± 0.00	**0.20** ± 0.01	0.10 ± 0.07
		6	0.00 ± 0.00	0.03 ± 0.02	**0.32** ± 0.03	0.26 ± 0.04	0.00 ± 0.01	0.00 ± 0.00	**0.32** ± 0.04	0.25 ± 0.07
		8	0.00 ± 0.01	0.05 ± 0.02	**0.47** ± 0.03	0.42 ± 0.05	0.01 ± 0.02	0.00 ± 0.00	**0.42** ± 0.07	0.38 ± 0.06
		10	0.01 ± 0.02	0.06 ± 0.03	**0.58** ± 0.05	0.54 ± 0.06	0.02 ± 0.02	0.00 ± 0.00	**0.57** ± 0.07	0.52 ± 0.07
	0.4	2	0.00 ± 0.00	0.01 ± 0.00	**0.08** ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.07** ± 0.01	0.01 ± 0.02
		4	0.00 ± 0.00	0.02 ± 0.01	**0.21** ± 0.02	0.13 ± 0.07	0.00 ± 0.00	0.00 ± 0.00	**0.21** ± 0.02	0.13 ± 0.07
		6	0.00 ± 0.01	0.04 ± 0.02	**0.35** ± 0.01	0.31 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	**0.36** ± 0.03	0.31 ± 0.03
		8	0.01 ± 0.02	0.05 ± 0.02	**0.52** ± 0.04	0.46 ± 0.05	0.00 ± 0.01	0.00 ± 0.00	**0.52** ± 0.03	0.47 ± 0.04
		10	0.01 ± 0.01	0.07 ± 0.03	**0.69** ± 0.04	0.66 ± 0.04	0.03 ± 0.02	0.00 ± 0.00	**0.69** ± 0.06	0.63 ± 0.07
	0.6	2	0.00 ± 0.00	0.01 ± 0.00	**0.07** ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.07** ± 0.01	0.00 ± 0.00
		4	0.00 ± 0.00	0.01 ± 0.01	**0.23** ± 0.01	0.16 ± 0.06	0.00 ± 0.00	0.00 ± 0.00	**0.21** ± 0.01	0.14 ± 0.06
		6	0.00 ± 0.01	0.04 ± 0.01	**0.38** ± 0.04	0.33 ± 0.03	0.01 ± 0.01	0.00 ± 0.00	**0.38** ± 0.02	0.33 ± 0.02
		8	0.01 ± 0.01	0.05 ± 0.02	**0.56** ± 0.05	0.50 ± 0.05	0.01 ± 0.01	0.00 ± 0.00	**0.55** ± 0.03	0.48 ± 0.05
		10	0.02 ± 0.02	0.08 ± 0.02	**0.75** ± 0.06	0.70 ± 0.05	0.02 ± 0.02	0.00 ± 0.00	**0.74** ± 0.05	0.68 ± 0.05
	0.8	2	0.00 ± 0.00	0.01 ± 0.00	**0.07** ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.08** ± 0.01	0.01 ± 0.02
		4	0.00 ± 0.00	0.02 ± 0.01	**0.22** ± 0.02	0.11 ± 0.08	0.00 ± 0.01	0.00 ± 0.00	**0.22** ± 0.01	0.15 ± 0.06
		6	0.00 ± 0.01	0.04 ± 0.01	**0.38** ± 0.03	0.34 ± 0.03	0.01 ± 0.01	0.00 ± 0.00	**0.38** ± 0.03	0.33 ± 0.05
		8	0.01 ± 0.01	0.05 ± 0.03	**0.57** ± 0.04	0.51 ± 0.05	0.02 ± 0.03	0.00 ± 0.01	**0.56** ± 0.03	0.50 ± 0.05
		10	0.02 ± 0.02	0.07 ± 0.03	**0.77** ± 0.05	0.72 ± 0.06	0.03 ± 0.02	0.00 ± 0.01	**0.79** ± 0.07	0.72 ± 0.06
	1.0	2	0.00 ± 0.00	0.01 ± 0.00	**0.07** ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**0.07** ± 0.01	0.00 ± 0.00
		4	0.00 ± 0.01	0.02 ± 0.01	**0.22** ± 0.01	0.15 ± 0.06	0.00 ± 0.01	0.00 ± 0.00	**0.22** ± 0.01	0.15 ± 0.06
		6	0.01 ± 0.01	0.05 ± 0.02	**0.38** ± 0.02	0.34 ± 0.03	0.01 ± 0.01	0.00 ± 0.00	**0.38** ± 0.02	0.34 ± 0.03
		8	0.02 ± 0.02	0.05 ± 0.02	**0.56** ± 0.02	0.51 ± 0.04	0.02 ± 0.02	0.01 ± 0.00	**0.56** ± 0.02	0.51 ± 0.04
		10	0.03 ± 0.03	0.08 ± 0.02	**0.78** ± 0.05	0.71 ± 0.08	0.03 ± 0.03	0.01 ± 0.01	**0.78** ± 0.05	0.71 ± 0.08

Table 7: Minimum group representation $\text{minrep}(C, S')$ **in the clusterings returned by the baselines and our algorithms, for the** k**-Means objective (all values of** k/t**). Results are reported as** mean ± standard deviation **over ten independent runs.**

		Global Representation ($\text{minrep}_{\text{global}}$)				Local Representation ($\text{minrep}_{\text{local}}$)			
Dataset	k/t	KM++	HDKM	PILP	ALP	KM++	HDKM	PILP	ALP
Heart	2	0.13 ± 0.11	0.19 ± 0.06	**0.32** ± 0.02	0.27 ± 0.04	0.20 ± 0.16	0.36 ± 0.13	**0.53** ± 0.05	0.50 ± 0.06
	4	0.17 ± 0.11	0.25 ± 0.06	**0.40** ± 0.02	0.36 ± 0.04	0.40 ± 0.24	0.99 ± 0.18	**1.36** ± 0.21	1.35 ± 0.22
	6	0.24 ± 0.08	0.31 ± 0.06	**0.44** ± 0.02	0.41 ± 0.04	0.94 ± 0.40	1.73 ± 0.15	**2.32** ± 0.35	2.27 ± 0.37
	8	0.25 ± 0.08	0.29 ± 0.04	**0.47** ± 0.01	0.43 ± 0.03	1.40 ± 0.42	2.36 ± 0.18	**3.02** ± 0.28	**3.02** ± 0.28
	10	0.27 ± 0.09	0.29 ± 0.04	**0.50** ± 0.02	0.48 ± 0.03	2.16 ± 0.58	3.29 ± 0.25	**4.06** ± 0.53	4.03 ± 0.52
Student	2	0.03 ± 0.05	0.22 ± 0.02	**0.28** ± 0.03	0.24 ± 0.05	0.08 ± 0.16	0.29 ± 0.02	**0.48** ± 0.05	0.47 ± 0.05
	4	0.10 ± 0.05	0.25 ± 0.02	**0.34** ± 0.02	0.31 ± 0.03	0.38 ± 0.25	0.66 ± 0.08	**1.18** ± 0.17	1.16 ± 0.17
	6	0.11 ± 0.08	0.28 ± 0.02	**0.36** ± 0.01	0.35 ± 0.02	0.71 ± 0.44	1.08 ± 0.17	**1.93** ± 0.30	1.92 ± 0.29
	8	0.11 ± 0.06	0.32 ± 0.02	**0.38** ± 0.01	0.36 ± 0.02	0.97 ± 0.60	1.66 ± 0.16	**2.71** ± 0.34	2.67 ± 0.35
	10	0.13 ± 0.04	0.32 ± 0.02	**0.39** ± 0.01	0.38 ± 0.02	1.33 ± 0.71	2.27 ± 0.24	**3.51** ± 0.34	3.49 ± 0.32
MovieLens	2	0.09 ± 0.07	0.13 ± 0.03	**0.30** ± 0.03	0.27 ± 0.05	0.18 ± 0.10	0.27 ± 0.03	**0.48** ± 0.05	0.44 ± 0.03
	4	0.11 ± 0.07	0.15 ± 0.01	**0.34** ± 0.01	0.30 ± 0.02	0.35 ± 0.16	0.53 ± 0.05	**1.06** ± 0.07	1.04 ± 0.07
	6	0.15 ± 0.06	0.18 ± 0.02	**0.35** ± 0.01	0.32 ± 0.02	0.57 ± 0.22	0.84 ± 0.04	**1.69** ± 0.11	1.66 ± 0.11
	8	0.15 ± 0.06	0.20 ± 0.02	**0.36** ± 0.01	0.34 ± 0.02	0.76 ± 0.28	1.27 ± 0.07	**2.32** ± 0.16	2.29 ± 0.17
	10	0.15 ± 0.05	0.18 ± 0.02	**0.36** ± 0.01	0.35 ± 0.02	0.96 ± 0.32	1.49 ± 0.05	**3.00** ± 0.17	2.98 ± 0.18
Credit Card	2	0.00 ± 0.01	0.04 ± 0.02	**0.12** ± 0.01	0.06 ± 0.03	0.01 ± 0.02	0.06 ± 0.01	**0.21** ± 0.01	0.20 ± 0.01
	4	0.01 ± 0.01	0.08 ± 0.01	**0.16** ± 0.01	0.12 ± 0.02	0.02 ± 0.03	0.14 ± 0.02	**0.50** ± 0.03	0.47 ± 0.02
	6	0.02 ± 0.02	0.09 ± 0.01	**0.17** ± 0.01	0.13 ± 0.02	0.05 ± 0.05	0.19 ± 0.02	**0.82** ± 0.07	0.79 ± 0.09
	8	0.02 ± 0.02	0.09 ± 0.01	**0.18** ± 0.00	0.15 ± 0.02	0.09 ± 0.07	0.27 ± 0.04	**1.18** ± 0.10	1.13 ± 0.09
	10	0.02 ± 0.02	0.09 ± 0.02	**0.19** ± 0.00	0.15 ± 0.02	0.15 ± 0.10	0.32 ± 0.02	**1.53** ± 0.10	1.49 ± 0.11
Bank	2	0.00 ± 0.00	0.12 ± 0.03	**0.19** ± 0.01	0.16 ± 0.02	0.00 ± 0.00	0.07 ± 0.01	**0.17** ± 0.00	0.15 ± 0.03
	4	0.01 ± 0.02	0.14 ± 0.01	**0.21** ± 0.00	0.19 ± 0.02	0.01 ± 0.02	0.15 ± 0.02	**0.40** ± 0.02	**0.40** ± 0.02
	6	0.01 ± 0.01	0.15 ± 0.02	**0.22** ± 0.00	0.20 ± 0.01	0.03 ± 0.03	0.23 ± 0.03	**0.65** ± 0.02	0.63 ± 0.03
	8	0.02 ± 0.01	0.14 ± 0.02	**0.23** ± 0.00	0.21 ± 0.01	0.06 ± 0.03	0.29 ± 0.04	**0.89** ± 0.03	0.84 ± 0.06
	10	0.02 ± 0.02	0.14 ± 0.02	**0.24** ± 0.00	0.22 ± 0.01	0.08 ± 0.04	0.39 ± 0.05	**1.15** ± 0.02	1.12 ± 0.03
Adult	2	0.00 ± 0.00	0.03 ± 0.01	**0.12** ± 0.00	0.08 ± 0.01	0.00 ± 0.00	0.01 ± 0.00	**0.07** ± 0.01	0.00 ± 0.00
	4	0.00 ± 0.00	0.04 ± 0.01	**0.15** ± 0.00	0.13 ± 0.01	0.00 ± 0.00	0.03 ± 0.01	**0.18** ± 0.01	0.11 ± 0.06
	6	0.00 ± 0.00	0.05 ± 0.01	**0.17** ± 0.00	0.14 ± 0.01	0.00 ± 0.00	0.06 ± 0.01	**0.31** ± 0.02	0.27 ± 0.02
	8	0.00 ± 0.00	0.05 ± 0.01	**0.18** ± 0.00	0.15 ± 0.01	0.00 ± 0.01	0.09 ± 0.02	**0.44** ± 0.03	0.39 ± 0.05
	10	0.00 ± 0.01	0.06 ± 0.01	**0.19** ± 0.00	0.17 ± 0.01	0.02 ± 0.02	0.13 ± 0.01	**0.60** ± 0.05	0.54 ± 0.05